

Contents of Module 3

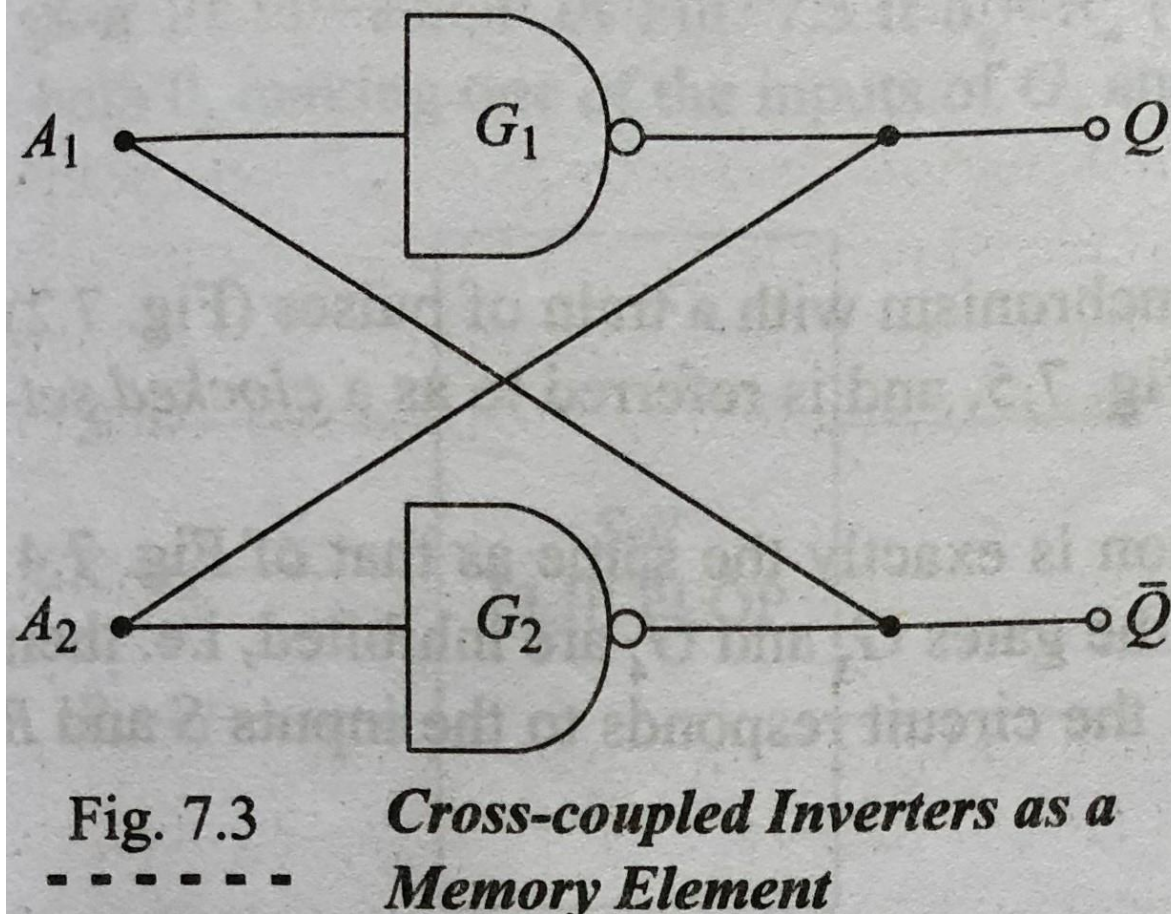
Unit No.	Contents	Hrs.
	Elements of Sequential Logic Design	
3	Sequential Logic: Latches and Flip-Flops. RS, JK, Master slave flip flops, T & D flip flops with various triggering methods, Conversion of flip flops. Counters: Asynchronous, Synchronous Counters, Up Down Counters, Mod Counters, Ring Counter, Twisted ring counter, Shift Registers, Universal Shift Register.	7

Text Book for module 3:

1

R. P. Jain, Modern Digital Electronics, Tata McGraw Hill Education, Third Edition 2003.

Flip-Flops (Memory Cells)



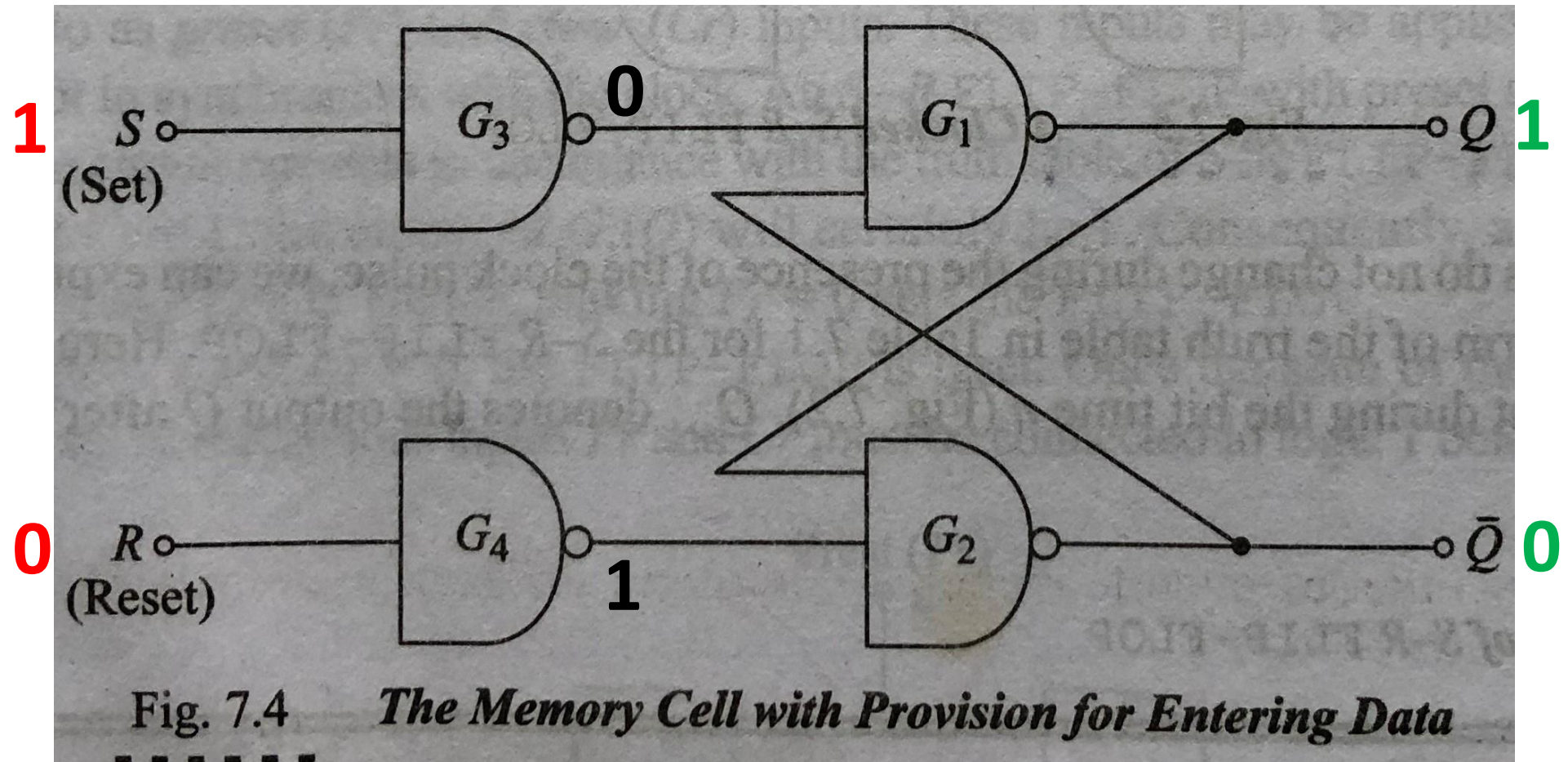
Cross coupled inverters as a memory element.

The basic memory circuit is known as FF. It has two stable states; logic high (1) (**Set state**) and logic low (0) (**Reset state**).

If $Q = 0$ then $\bar{Q} = 1$ and
If $Q = 1$ then $\bar{Q} = 0$
The outputs Q and $\bar{Q}$ are **complementary**.

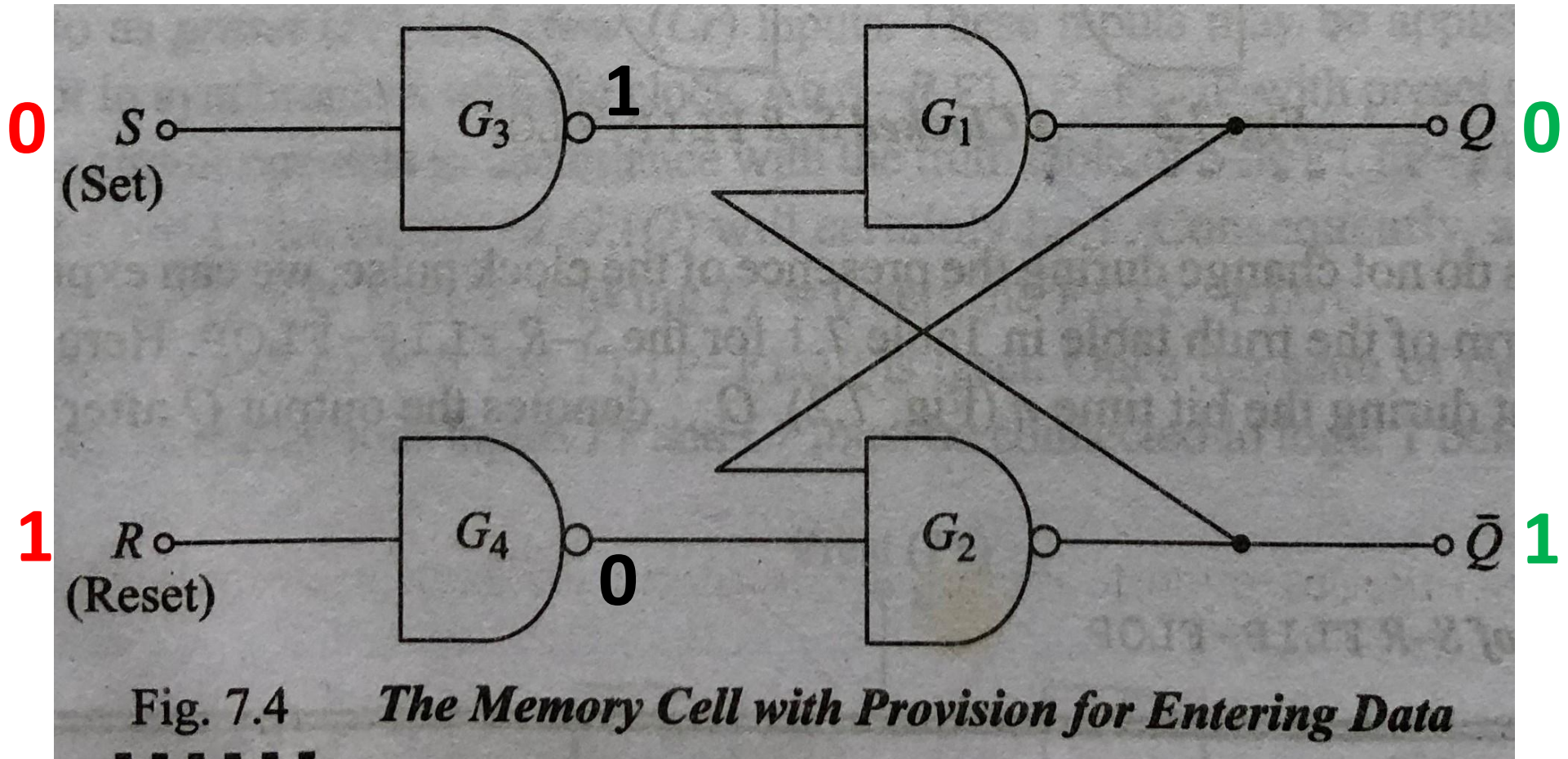
Since the information is locked or latched in the circuit, it is also called as a **Latch**

FF with provision for entering data



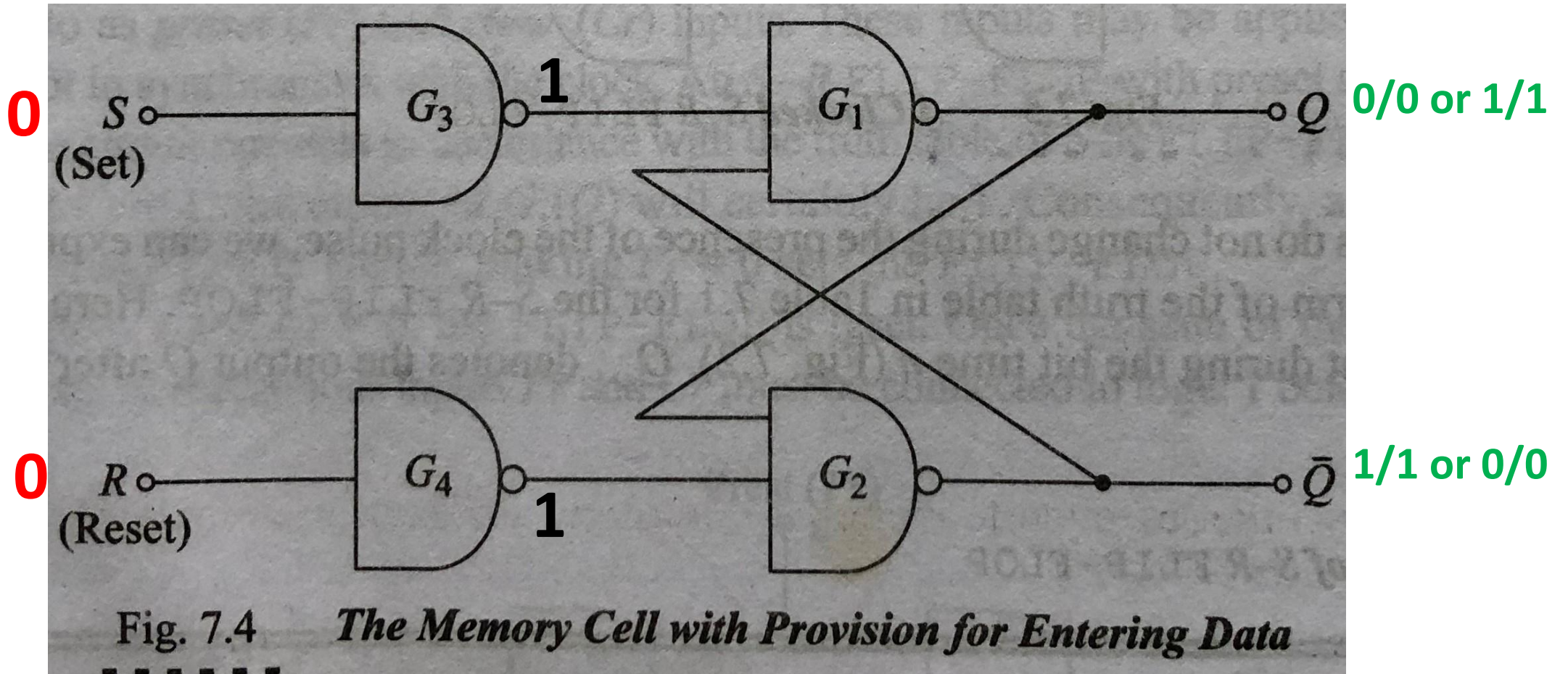
$S = 1, R = 0$ gives output $Q = 1$ and $\bar{Q} = 0$; This is Set state.

FF with provision for entering data



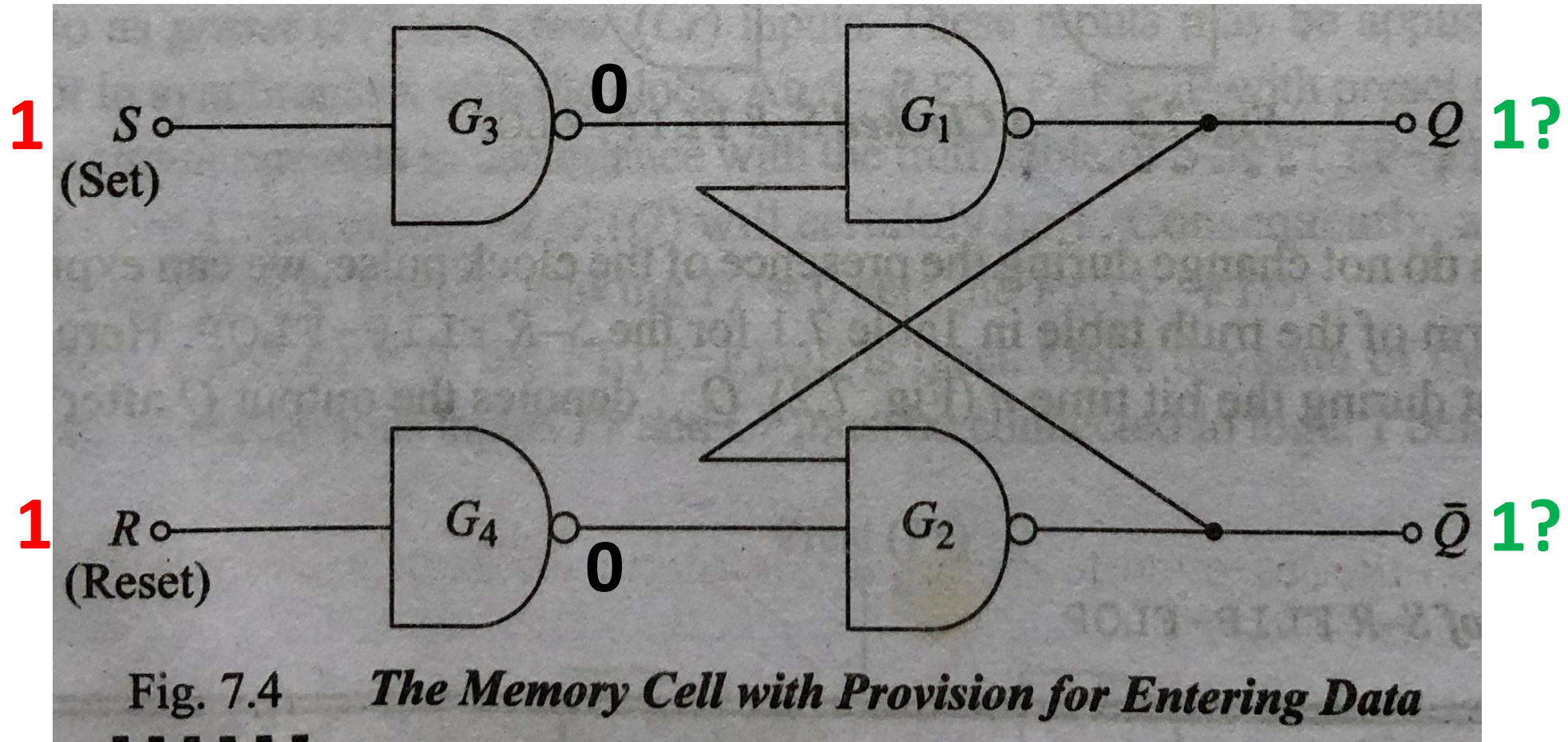
$S = 0$, $R = 1$ gives output $Q = 0$ and $Q_{\text{bar}} = 1$; This is Reset state or Clear state.

FF with provision for entering data



$S = 0, R = 0$ then Q and $Qbar$ outputs will remain as previous outputs, they will not change.

FF with provision for entering data



S = 1, R = 1 then both outputs are 1, this is forbidden state in SR FF.

Truth Table of SR Flip-Flop

Inputs		Output	
S	R	Present State Q_n	Next State Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1 ?
1	1	1	1 ?

Reduced Truth Table of SR Flip-Flop

Inputs		Output	Comment
S	R	Q_{n+1}	
0	0	Q_n	No Change
0	1	0	Reset
1	0	1	Set
1	1	?	Forbidden State

Derivation of Excitation Table of SR Flip-Flop

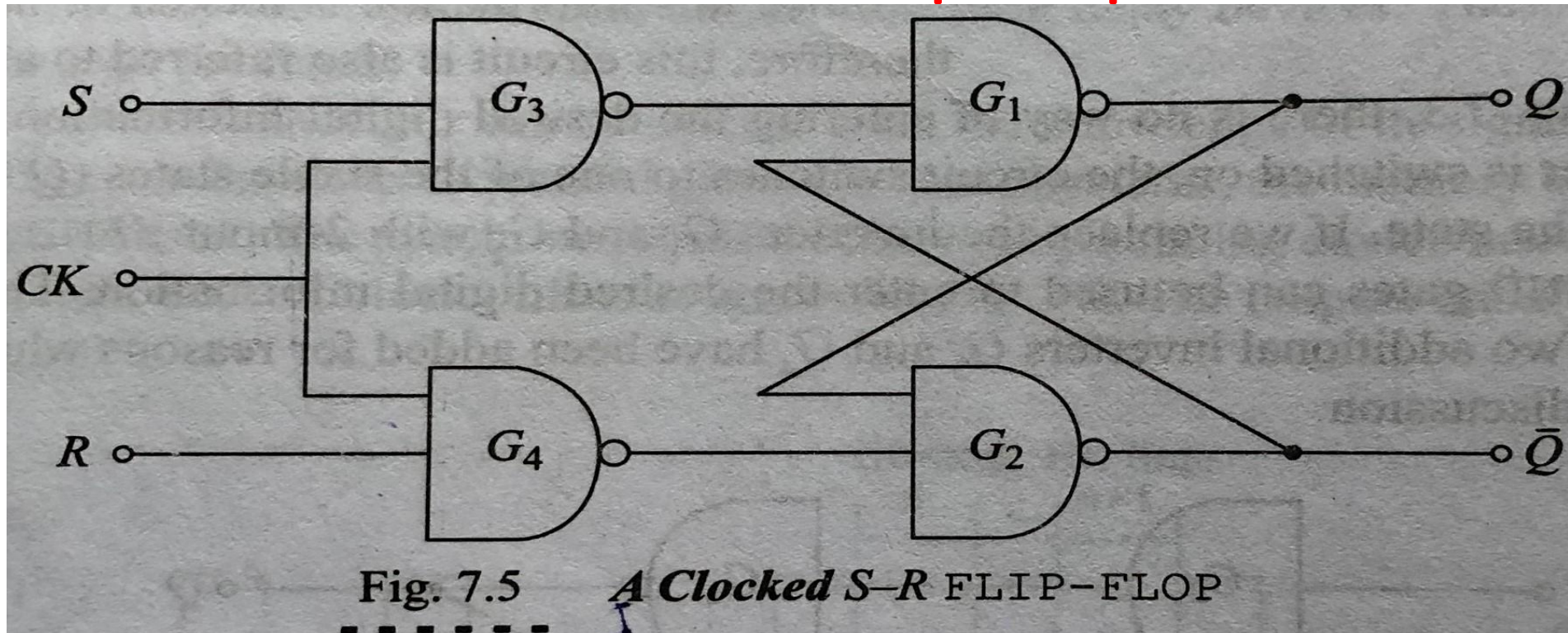
Inputs		Output	
S	R	P/S Q_n	N/S Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1 ?
1	1	1	1 ?

Excitation Table of SR FF

Output		Inputs	
Q_n	Q_{n+1}	S	R
0	0	0	d
0	1	1	0
1	0	0	1
1	1	d	0

Excitation Table gives all possibilities of present state, next state and the required inputs to have such change from present state to next state.

Clocked S-R Flip-Flop



When $\text{Clk} = 1$; gates G_3 , G_4 are enabled and its operation is just like previous circuit without clock provision.

When $\text{Clk} = 0$; gates G_3 , G_4 are disabled and its outputs are 1 irrespective of the values of S and R . In other words, the circuit responds to S and R inputs only when the clock is **High (1)**

Symbol of SR FF

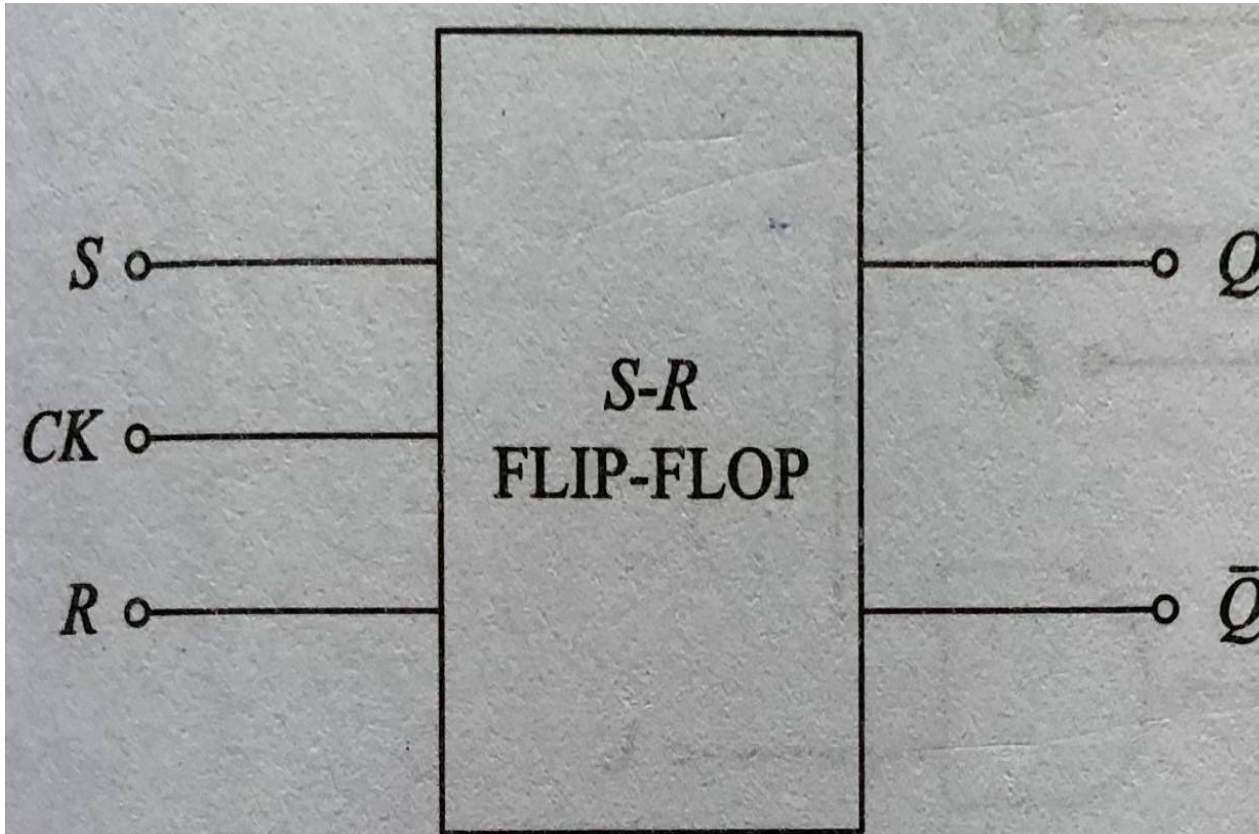
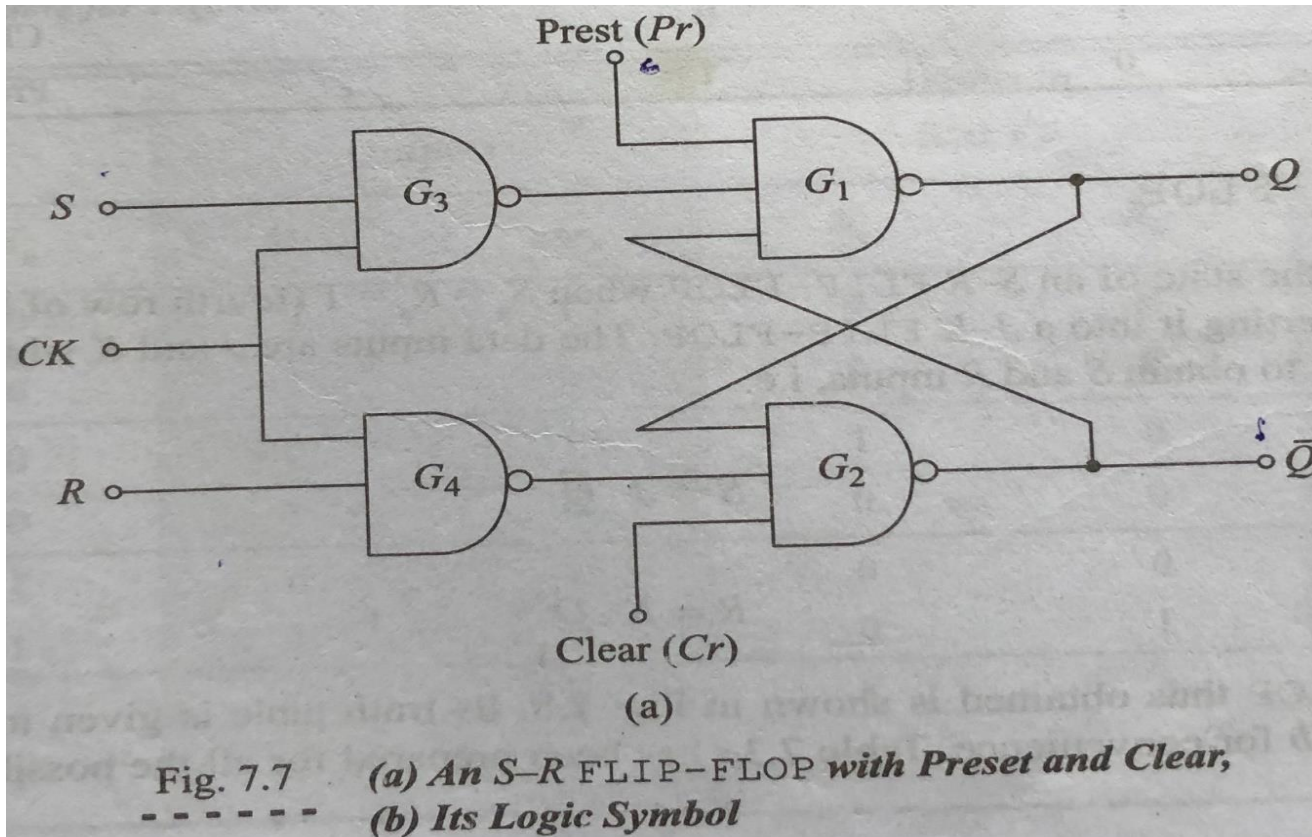


Fig. 7.6

Logic Symbol of Clocked S-R

FLIP-FLOP

SR FF with Preset and Clear pins



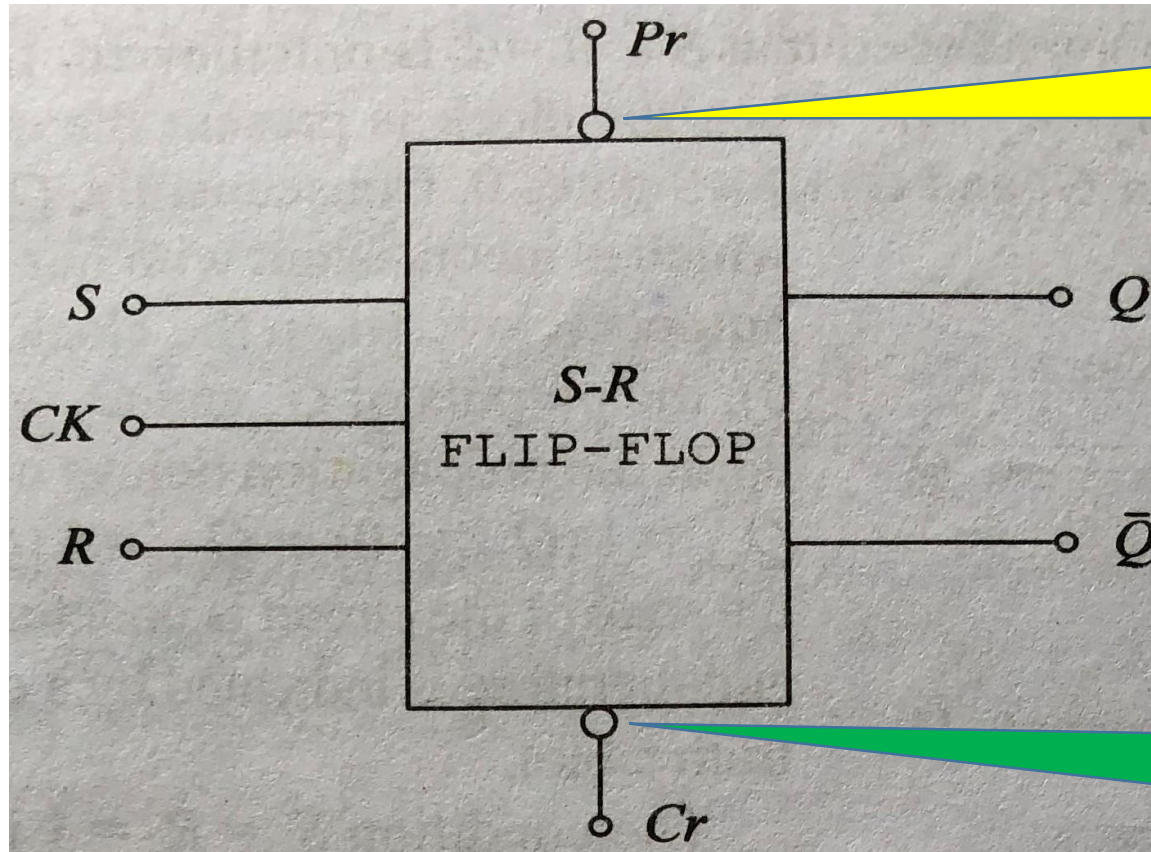
IF preset is Low and Clear is High; output Q will be High and Qbar will be Low.

IF preset is High and Clear is Low; output Q will be Low and Qbar will be High.

Both Preset and Clear should not be Low simultaneously because in such case $Q = \bar{Q} = 1$ (which is forbidden State)

Preset and Clear pins are active low means they will perform their intended function only when logic low level is applied to these pins. These pins will be disabled when connected to High level

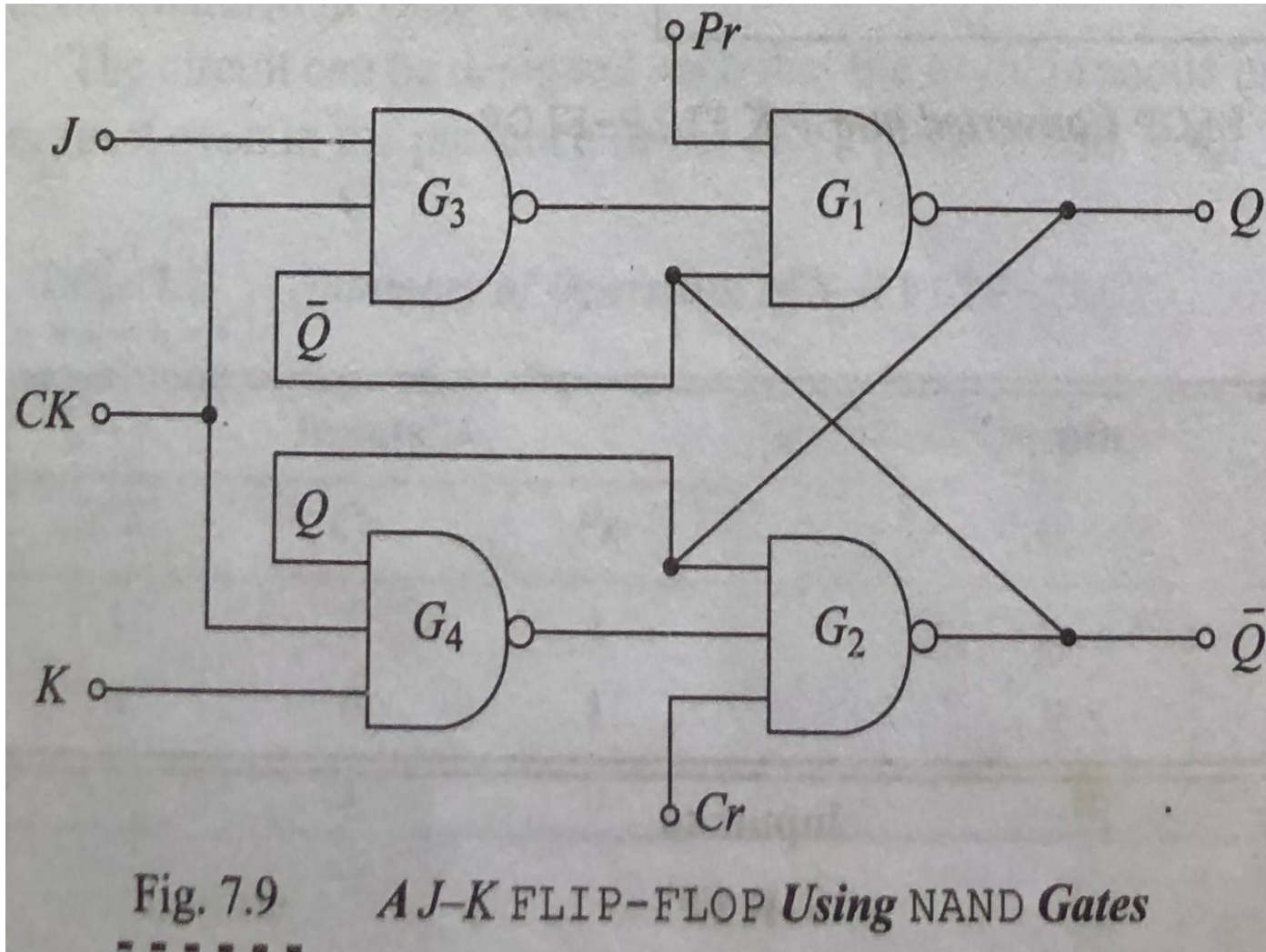
Symbol of SR FF with Preset and Clear pins



Active low Preset pin is indicated by bubble.

Active low Clear pin is indicated by bubble.

JK FF Using NAND Gates



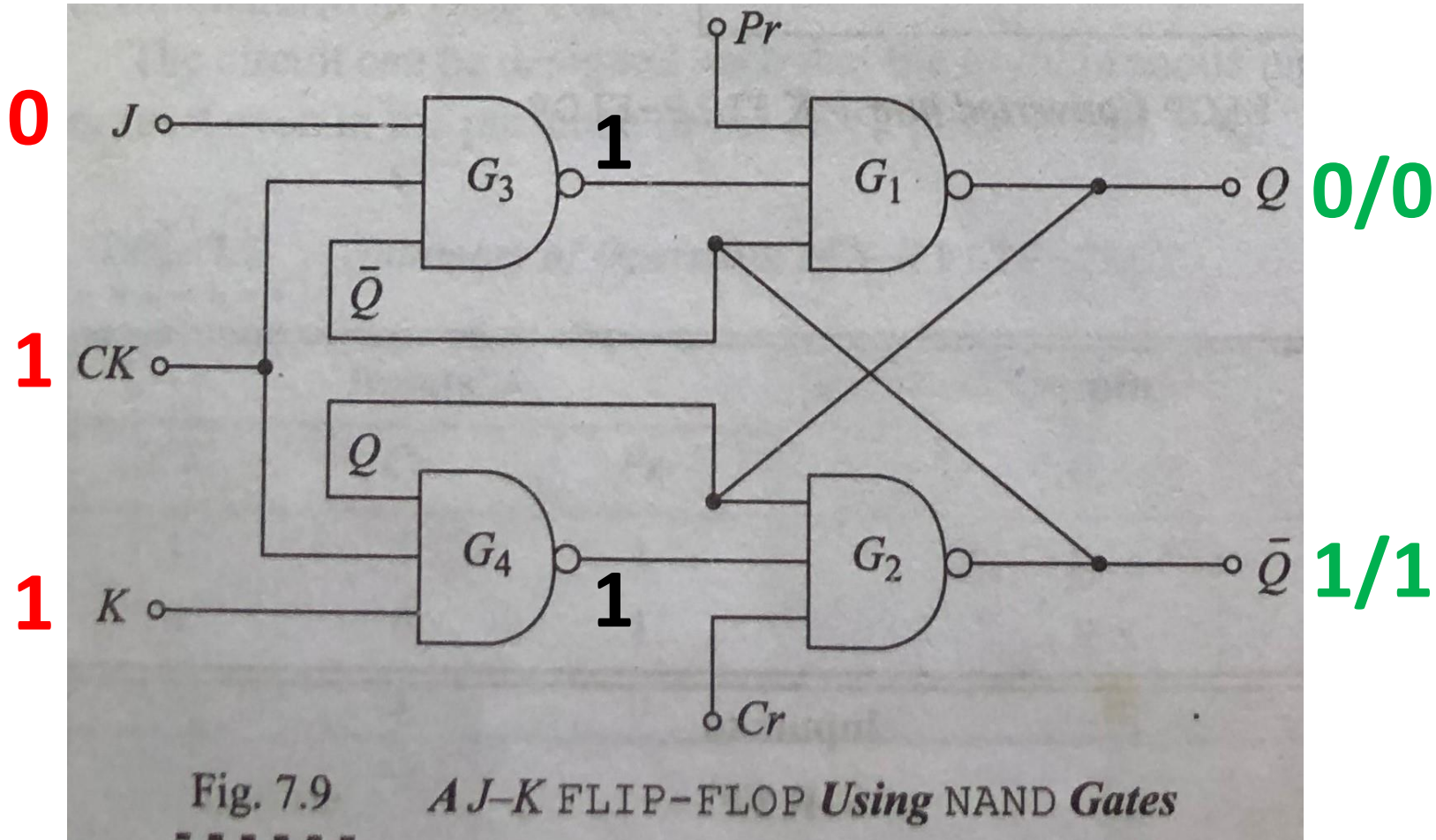
The difficulty of both inputs 1 ($S = R = 1$) being not allowed in SR FF is eliminated in JK FF by using the feedback connection from outputs to the inputs of the gates G_3 and G_4 .

0



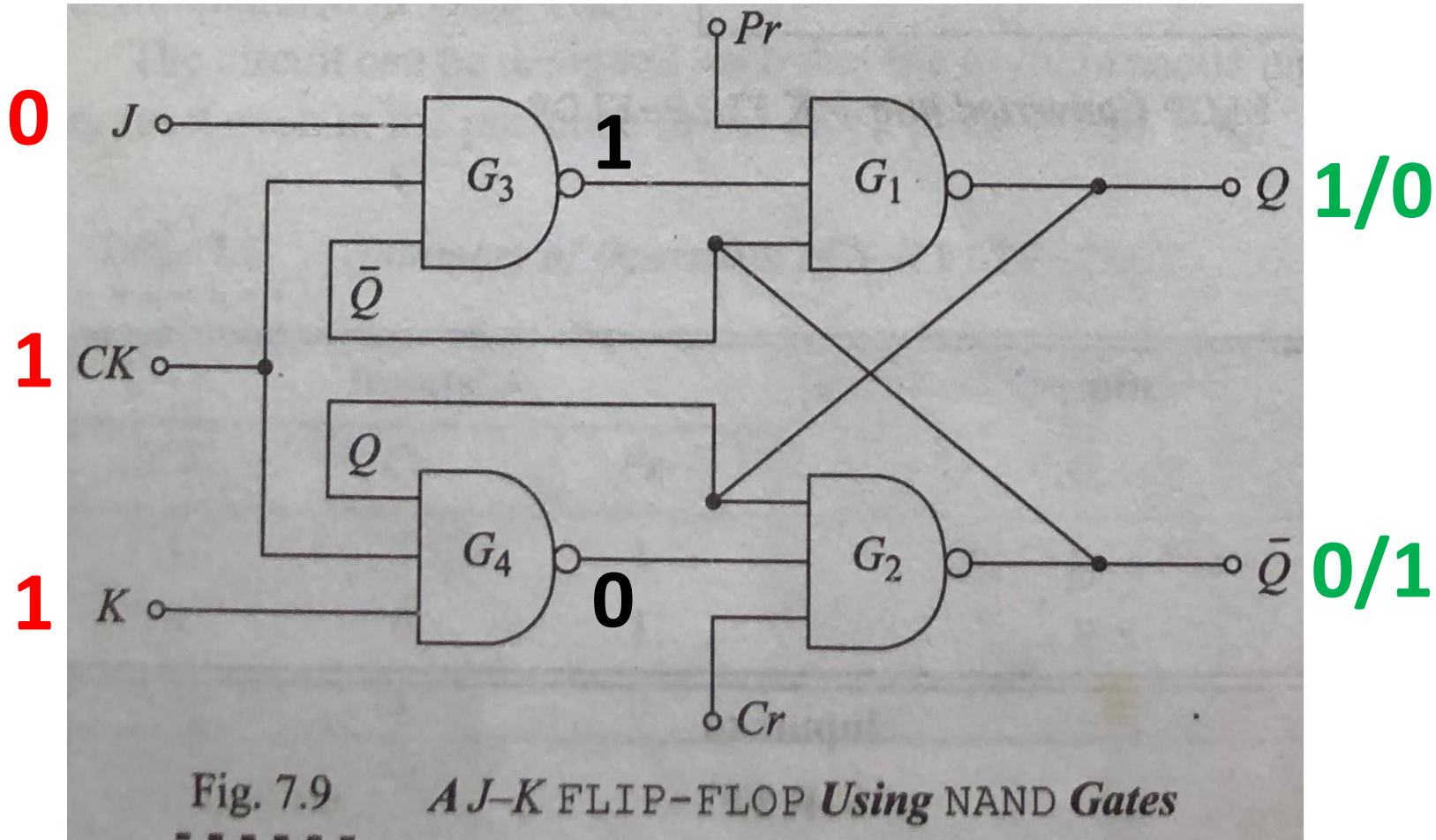
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JK FF Using NAND Gates



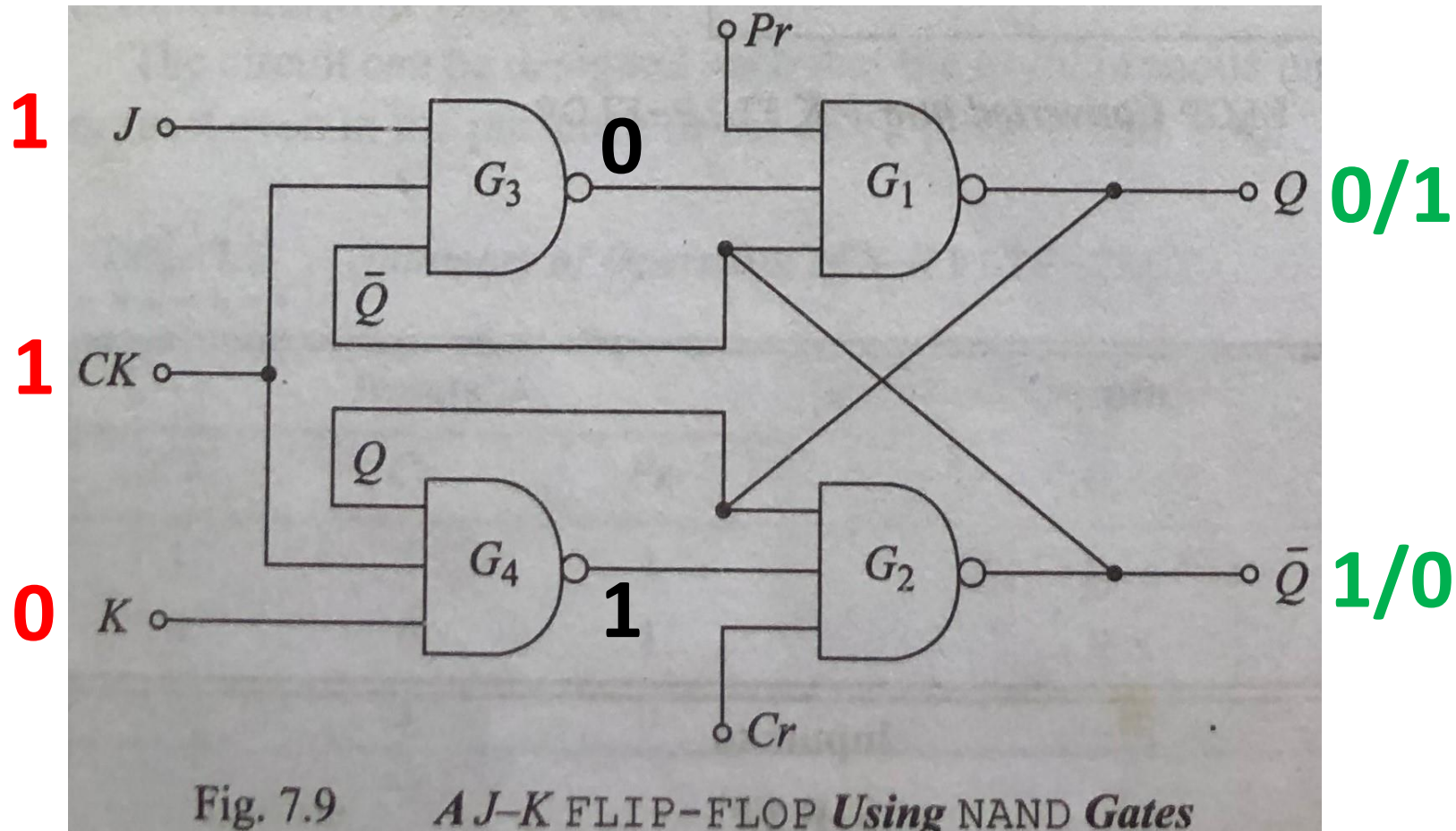
When $J = 0$ and $K = 1$, If Present Output is $Q = 0$, In next state also it will remain 0

JK FF Using NAND Gates



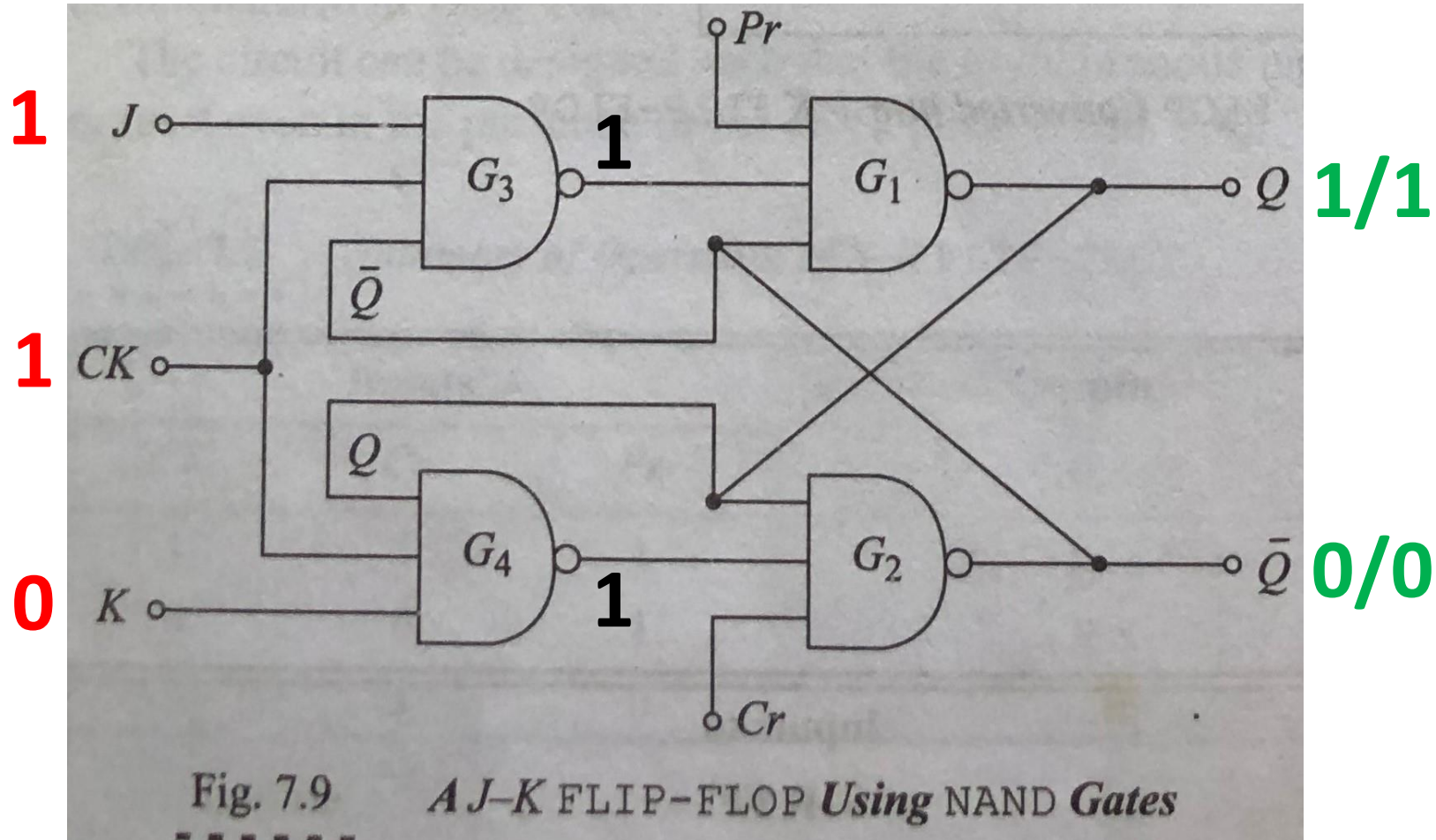
When $J = 0$ and $K = 1$, If Present Output is $Q = 1$, In next state it will become 0

JK FF Using NAND Gates



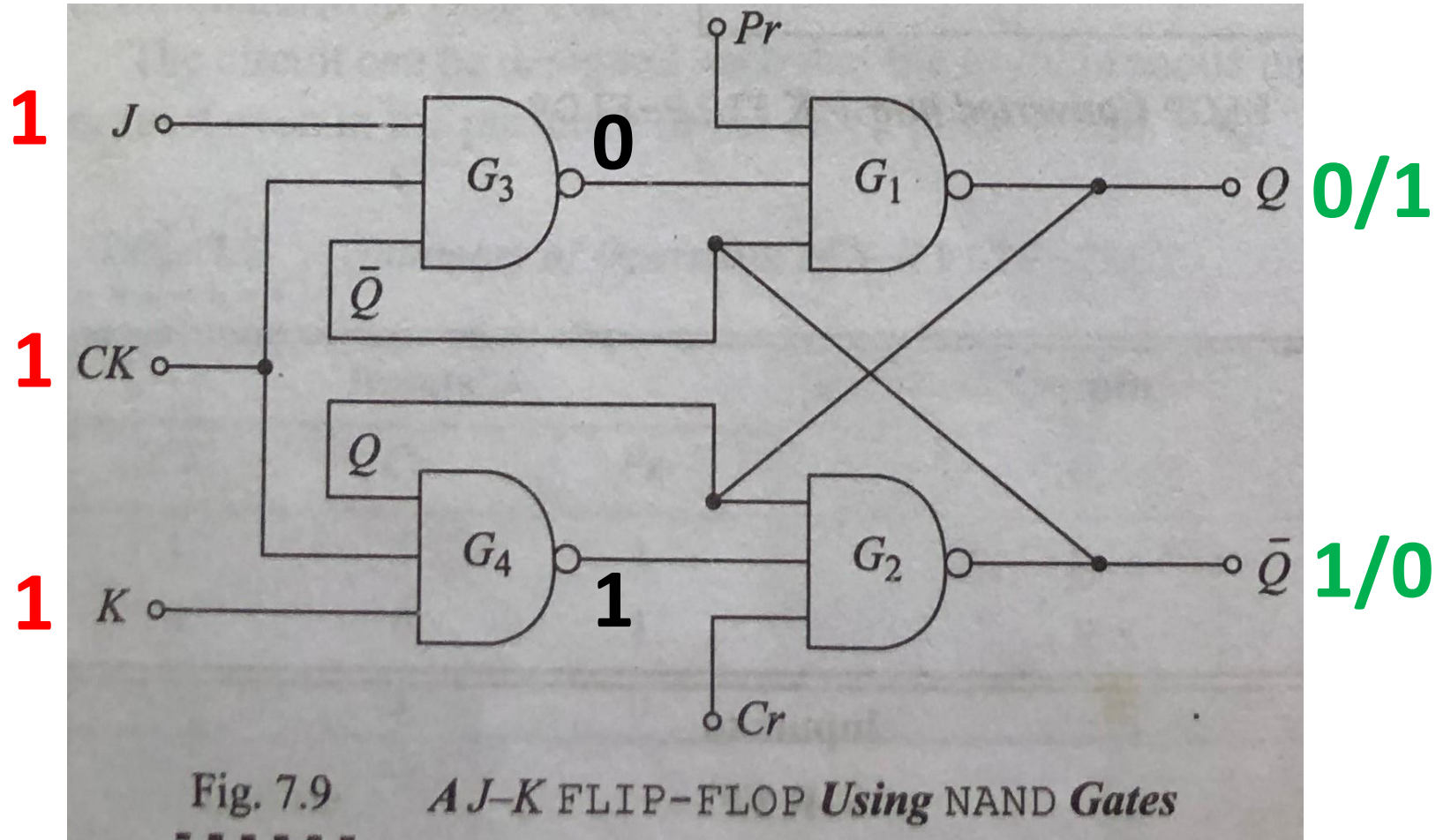
When $J = 1$ and $K = 0$, If Present Output is $Q = 0$, In next state it will become 1

JK FF Using NAND Gates



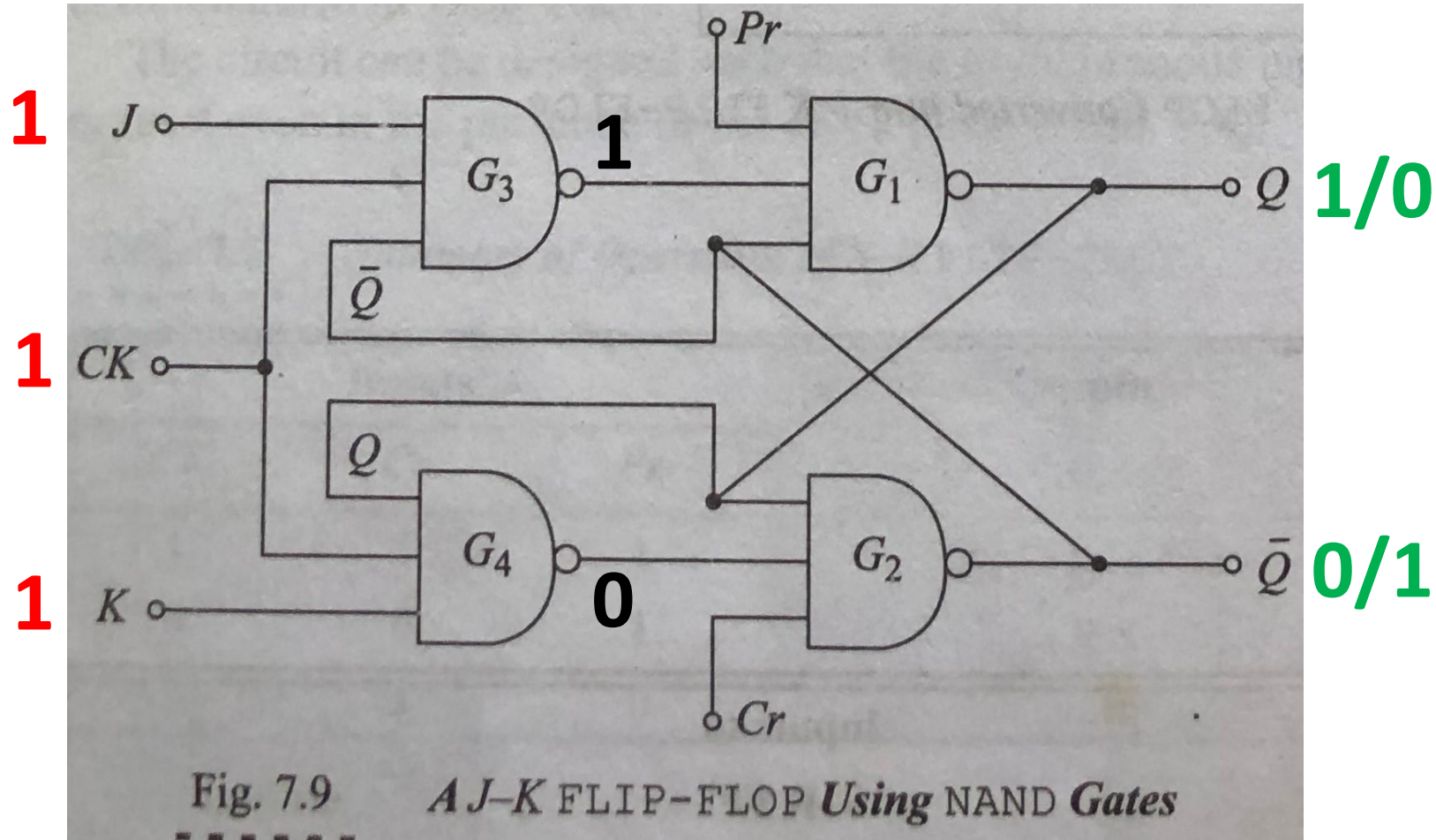
When $J = 1$ and $K = 0$, If Present Output is 1, In next state also it will remain 1

JK FF Using NAND Gates



When $J = K = 1$ and If Present Output is $Q = 0$, In next state it will change to 1

JK FF Using NAND Gates



When $J = K = 1$ and If Present Output is $Q = 1$, In next state it will change to 0

Truth Table of JK Flip-Flop

Inputs		Output	
J_n	K_n	Present State Q_n	Next State Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Reduced Truth Table of JK Flip-Flop

Inputs		Output	Comment
J_n	K_n	Q_{n+1}	
0	0	Q_n	No Change
0	1	0	Reset
1	0	1	Set
1	1	$\overline{Q_n}$	Toggle

Derivation of Excitation Table of JK Flip-Flop

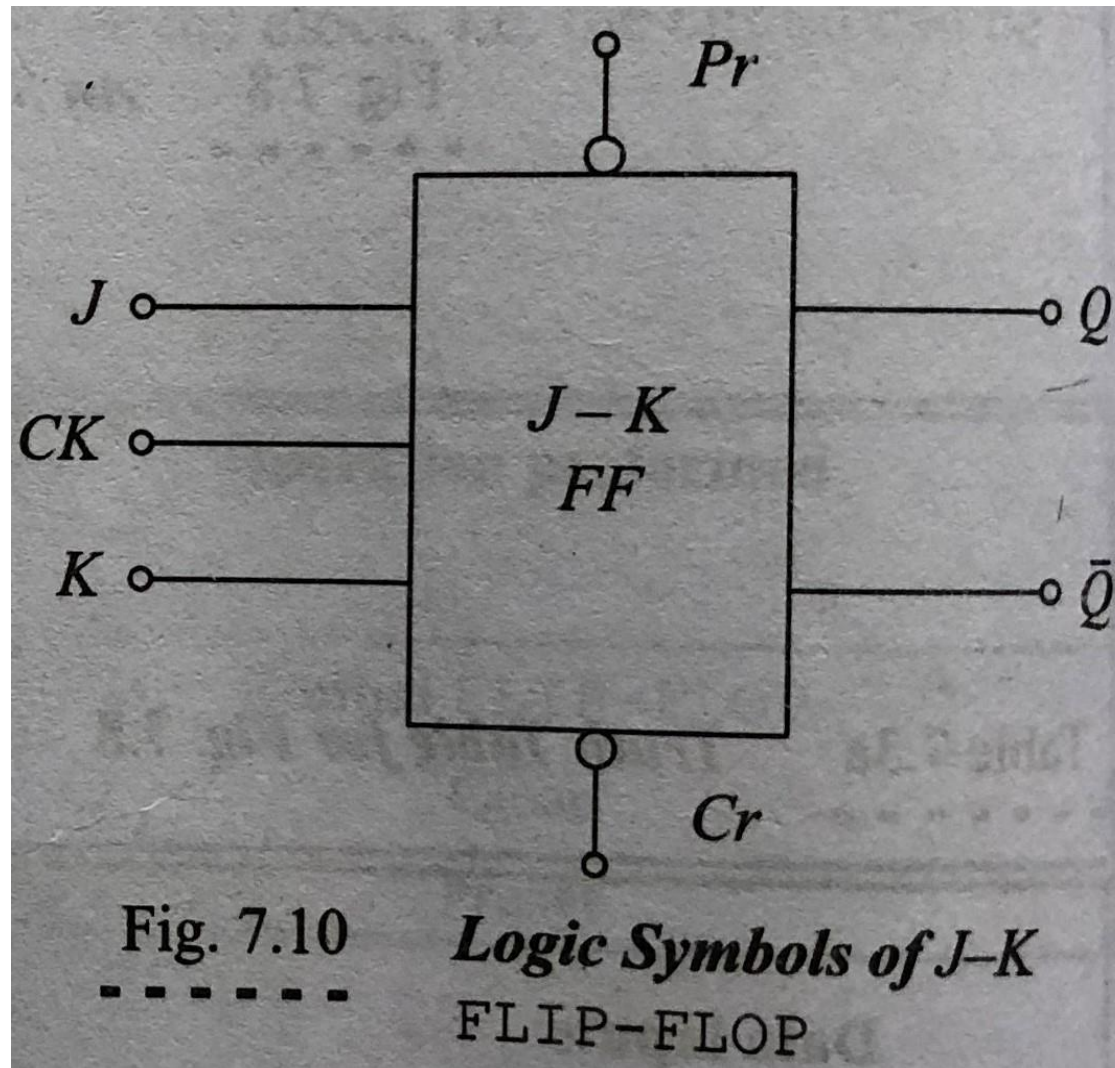
Inputs		Output	
J	K	P/S Q_n	N/S Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Excitation Table of JK FF

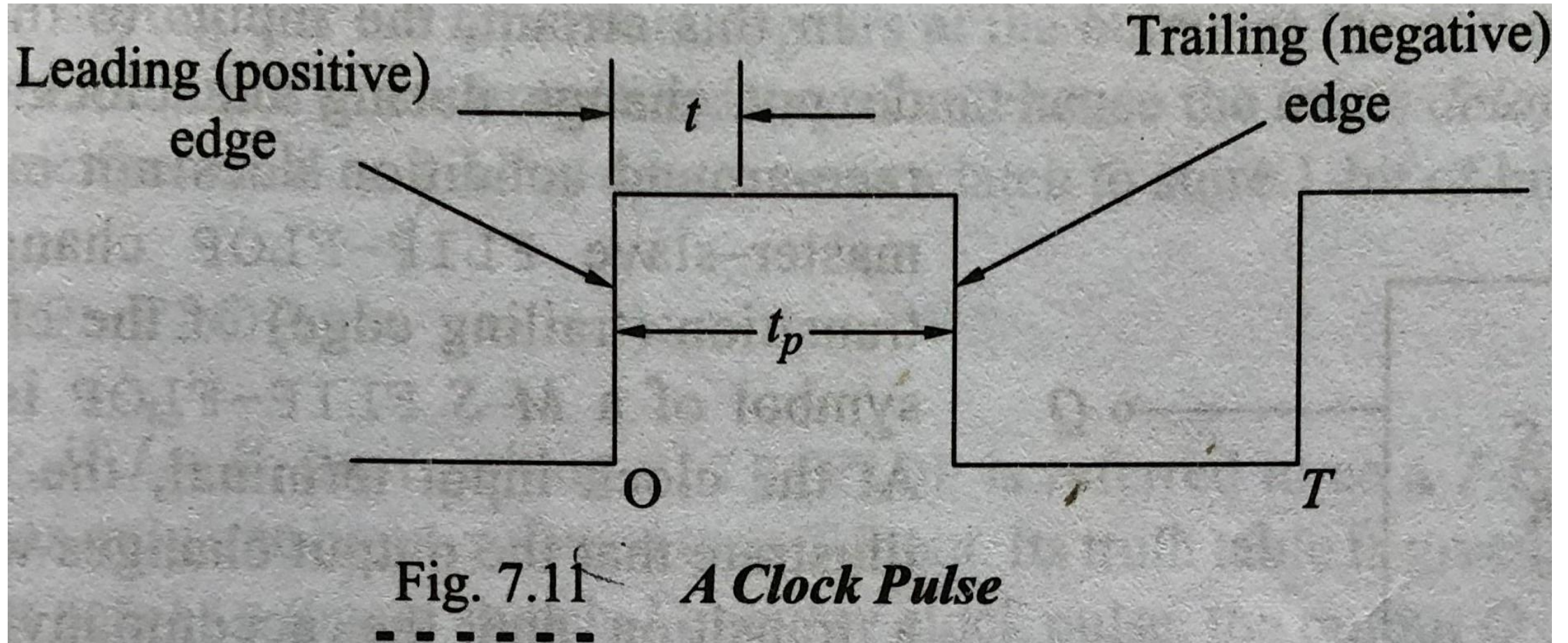
Output		Inputs	
Q_n	Q_{n+1}	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

Excitation Table gives all possibilities of present state, next state and the required inputs to have such change from present state to next state.

Logic Symbol of JK Flip-Flop



Race-Around Condition in JK FF



What is Race-around condition in JK FF? How can it be avoided?

If $J = K = 1$ and $Q = 0$ and clock pulse is as shown in fig, after a time interval Δt equal to the propagation delay through two NAND gates in series, the output will change to $Q = 1$.

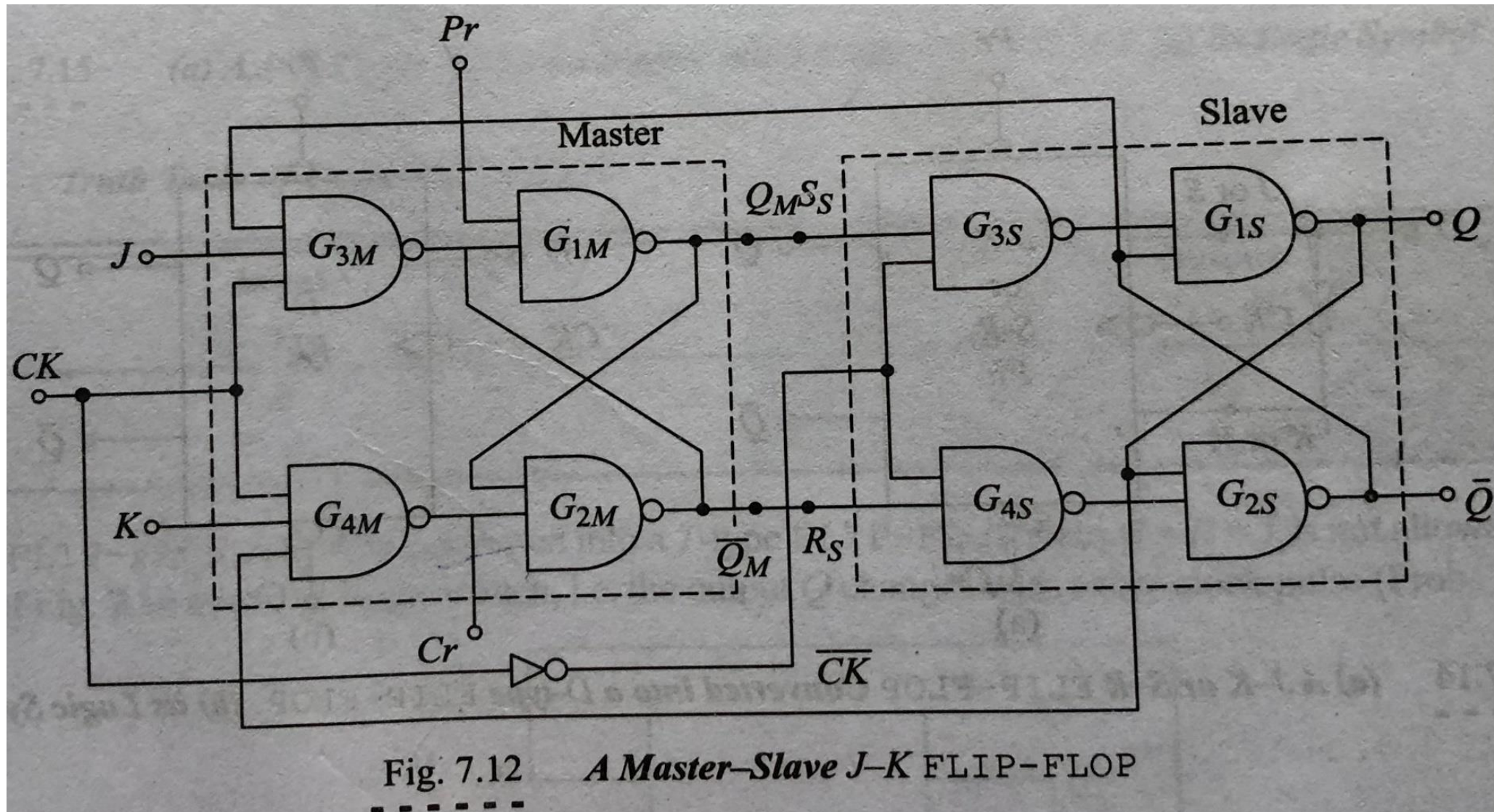
Now we have $J = K = 1$ and $Q = 1$ and after another time interval of Δt the output will change back to $Q = 0$.

Hence for the duration t_p of the clock pulse, the output will oscillate back and forth between 0 and 1.

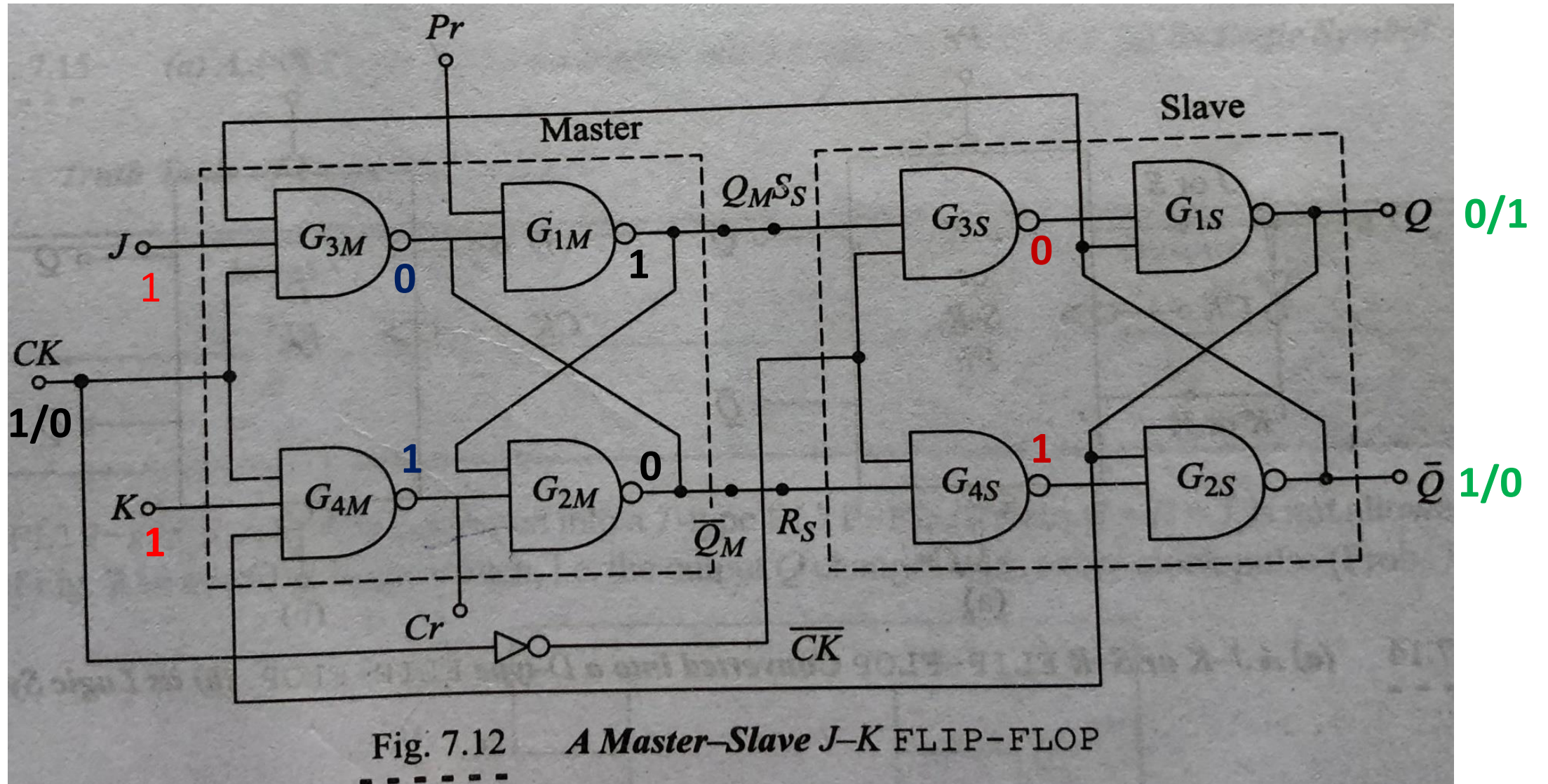
At the end of the clock pulse, the value of Q is uncertain. This situation is referred to as the ***Race-around condition***.

The race-around condition can be avoided if $t_p < \Delta t < T$. However, it may be difficult to satisfy this condition because of very small propagation delays in ICs. But this is done by using the edge-triggered flip flop rather than using the level-triggered flip-flop. A more practical method for overcoming this difficulty is the use of master-slave (M-S) configuration.

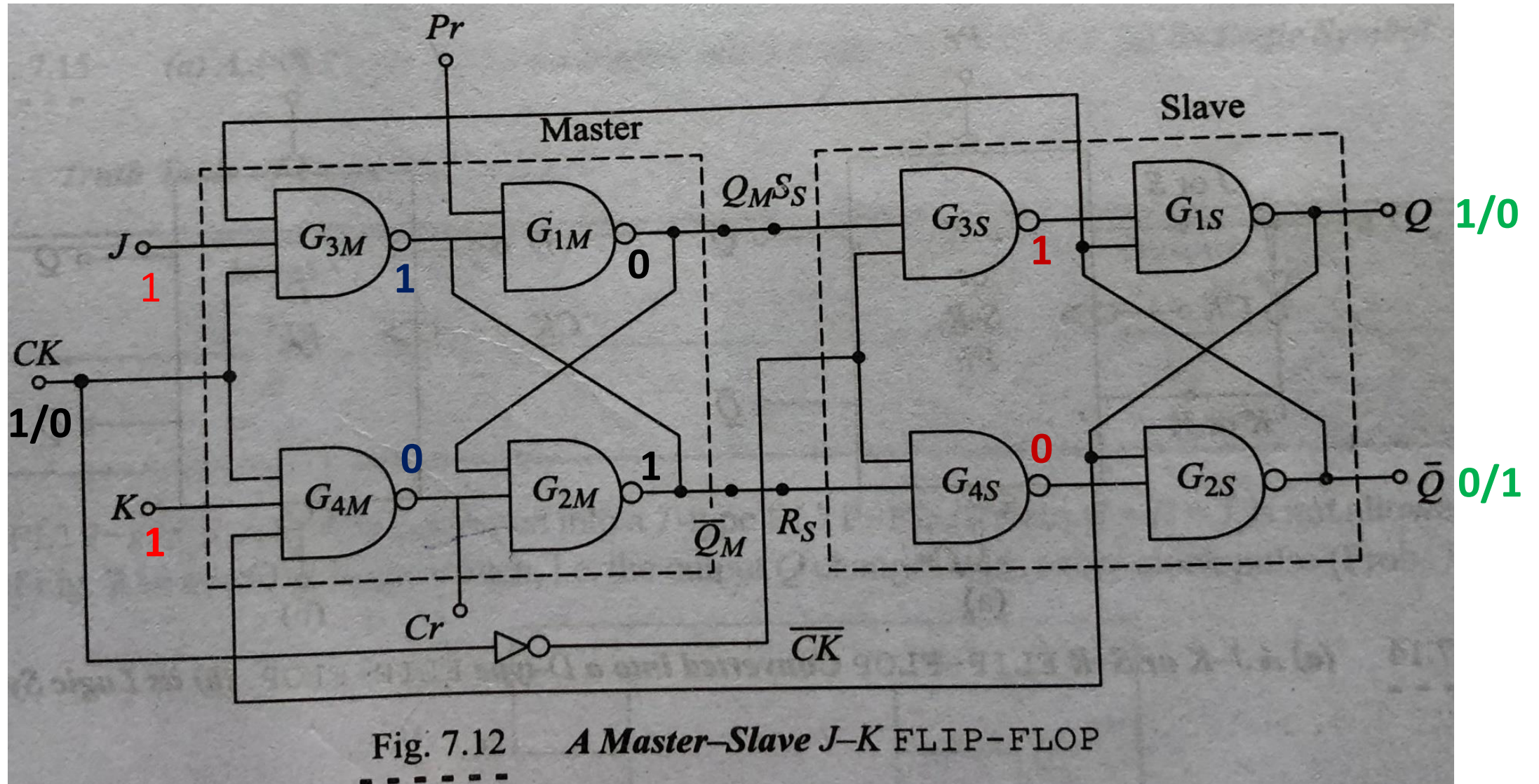
Master-Slave JK Flip-Flop



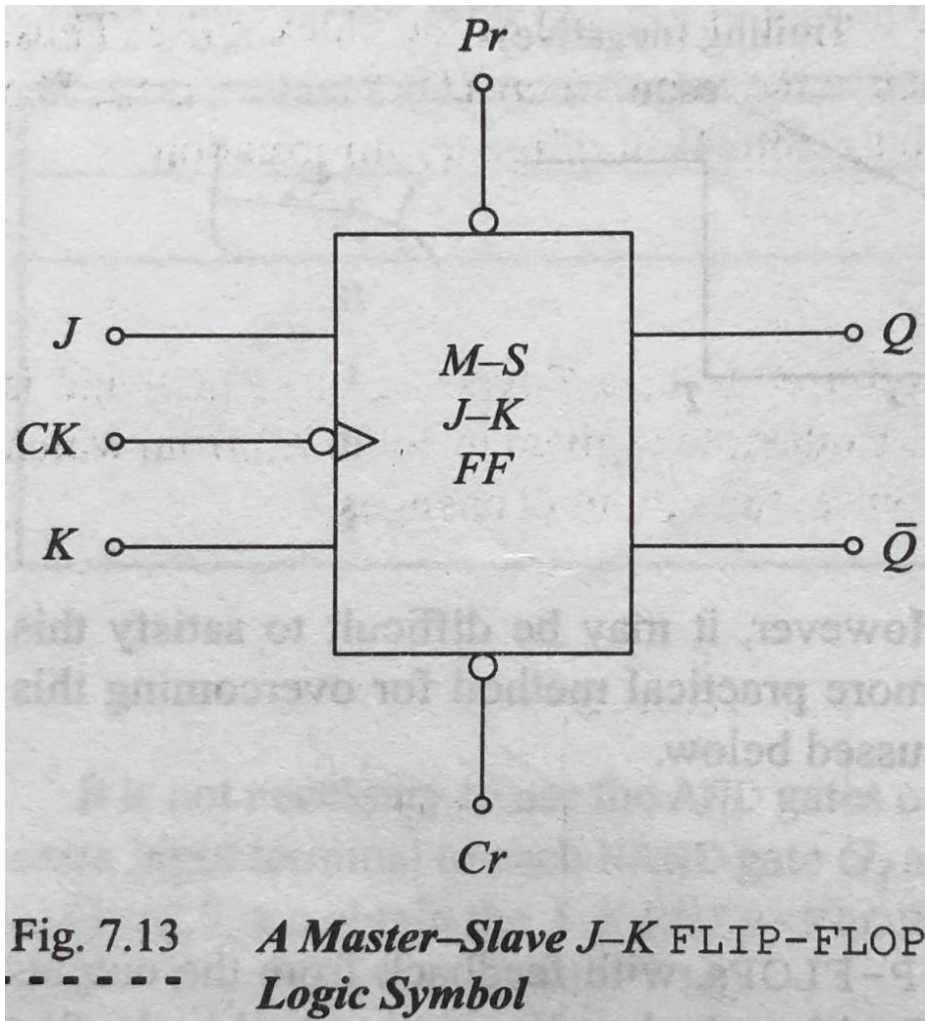
Master-Slave JK Flip-Flop



Master-Slave JK Flip-Flop



Toggle condition in JKMS Flip-Flop

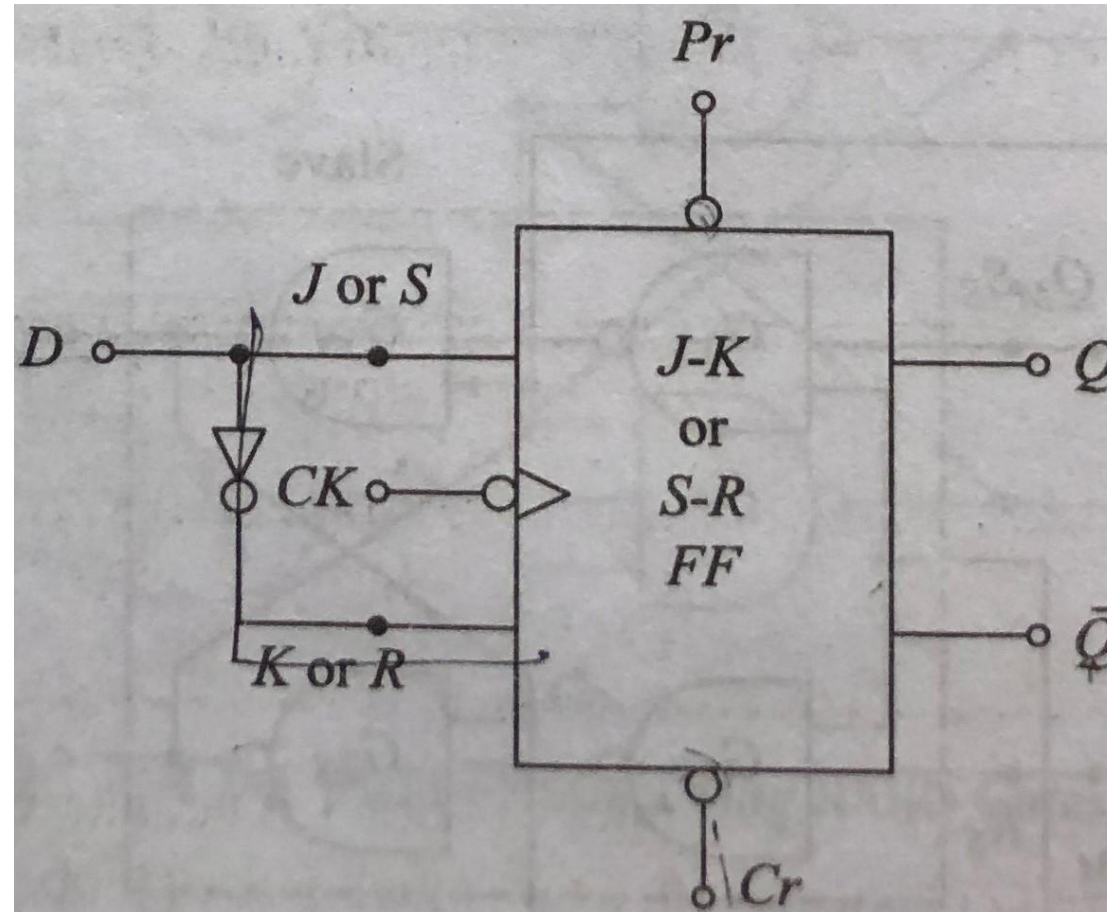


For $J = K = 1$, when clock goes from 1 to 0, output Q changes from 0 to 1 and 1 to 0 for every negative going edge of clock. This is called toggle condition of JKMS FF.

In JKMS FF the output of master stage changes for positive going edge of clock and output of slave changes for negative going edge of clock. The final output changes after negative going edge of clock, hence it is negative edge triggered FF, which is indicated by bubble at the clock input pin.

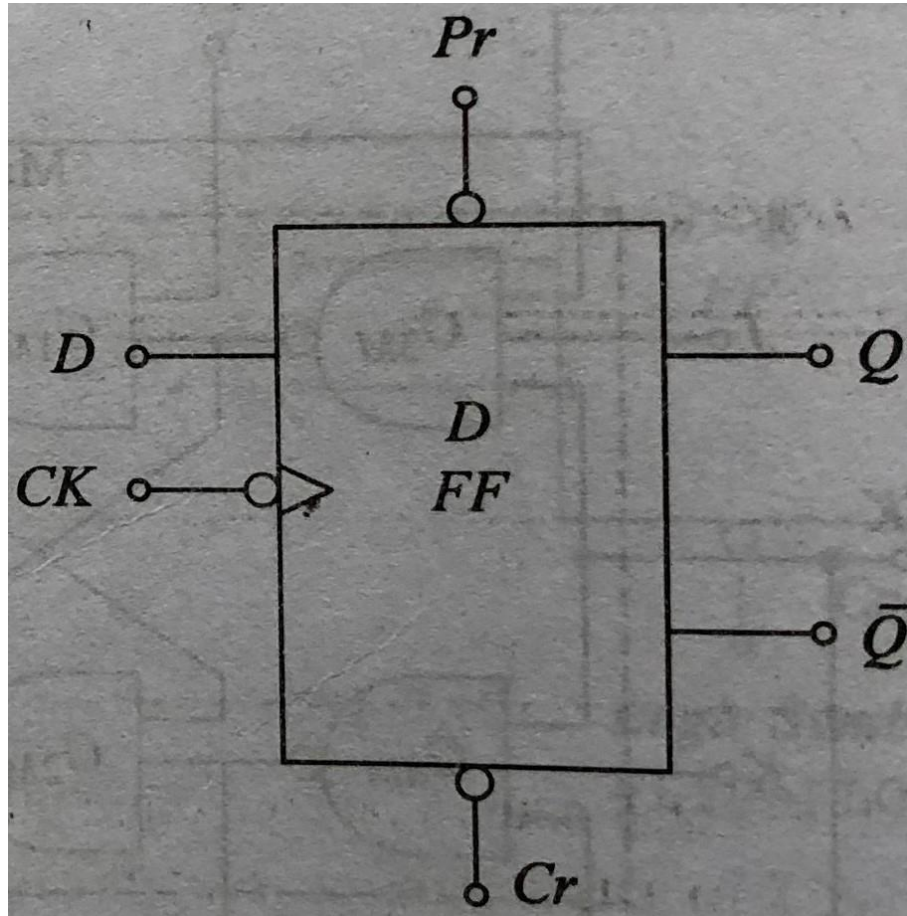
The second stage copies the action of first stage after negative going edge of clock, hence first stage is called master and second stage as slave.

D-Type Flip-Flop



Logic symbol of D Flip-Flop

TT of D FF



Input D_n	Output Q_{n+1}
0	0
1	1

In D FF Next State Output is Same as D

Derivation of Excitation Table of D Flip-Flop

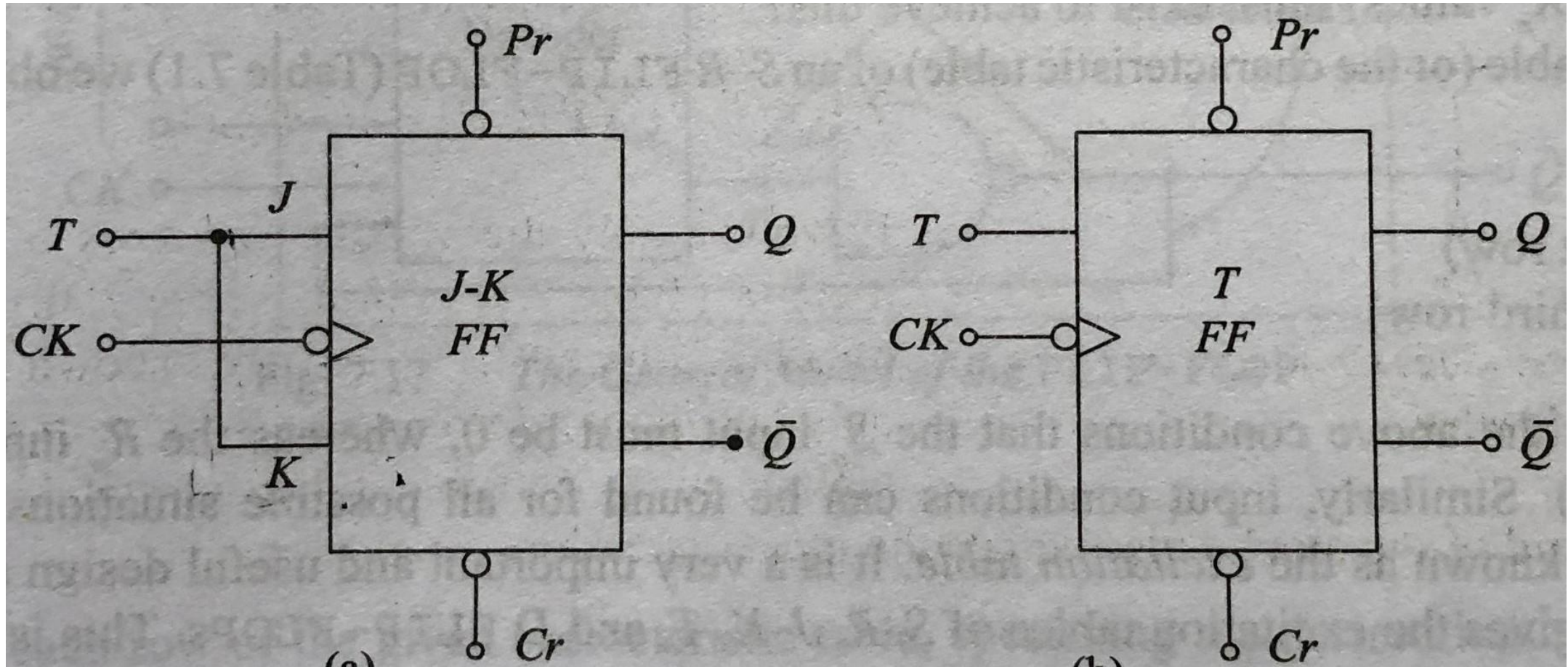
Input	Output	
D	P/S Q_n	N/S Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table of D FF

Output		Input
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table gives all possibilities of present state, next state and the required inputs to have such change from present state to next state.

T-Type Flip-Flop



JK FF converted to T FF

Logic Symbol of T FF

TT of T FF

Input T_n	Output Q_{n+1}
0	Q_n
1	$\overline{Q_n}$

When $T = 1$, For each clock pulse output is complement of previous state. It acts as a Toggle switch, because of this it is called as T (Toggle) FF.

Derivation of Excitation Table of T Flip-Flop

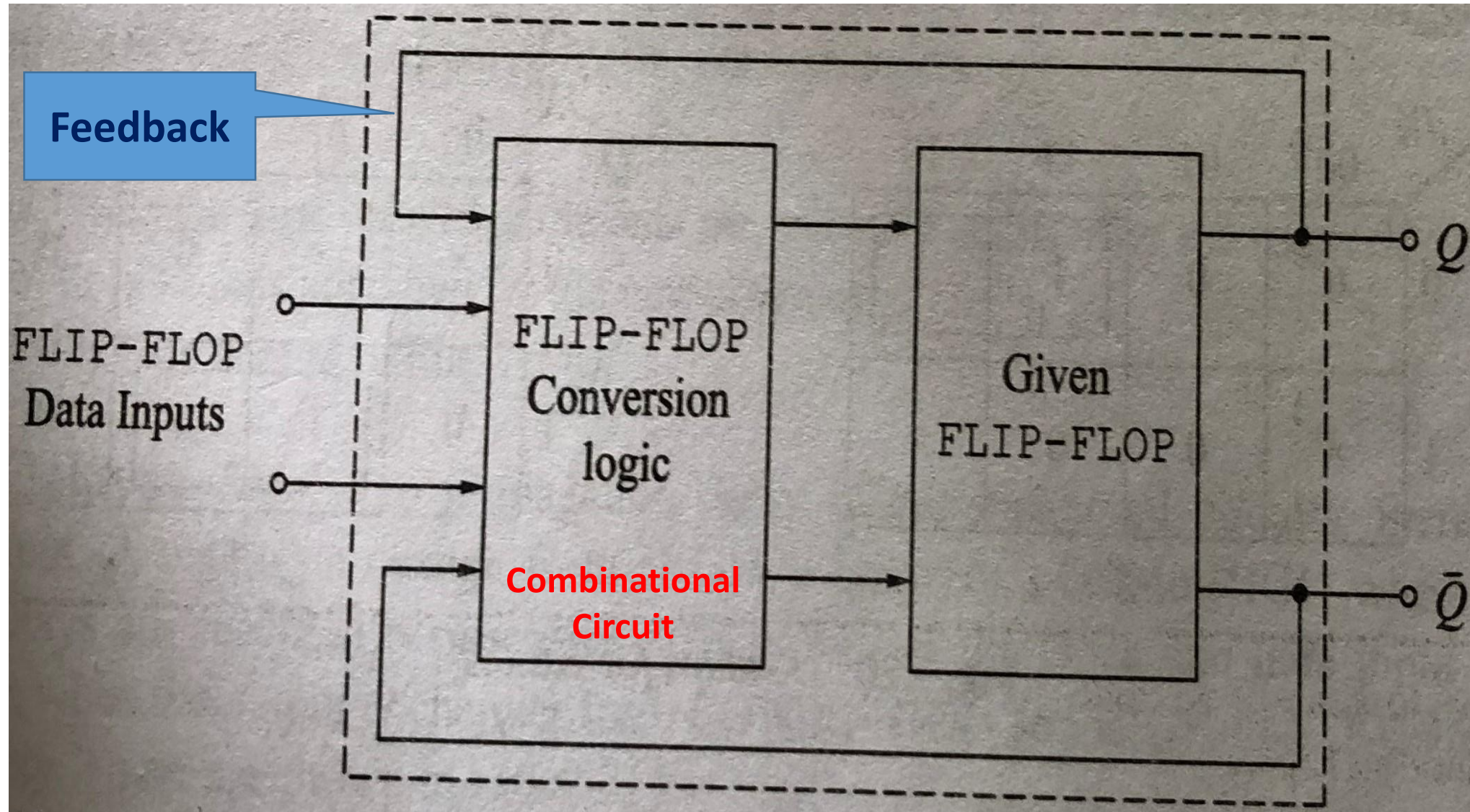
Input	Output	
T	P/S Q_n	N/S Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table of T FF

Output		Input
Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table gives all possibilities of present state, next state and the required inputs to have such change from present state to next state.

Conversion from One Type of Flip-Flop to Another Type



Reduced Truth Table of SR Flip-Flop

Inputs		Output	Comment
S	R	Q_{n+1}	
0	0	Q_n	No Change
0	1	0	Reset
1	0	1	Set
1	1	?	Forbidden State

Derivation of Excitation Table of SR Flip-Flop

Inputs		Output	
S	R	P/S Q_n	N/S Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1 ?
1	1	1	1 ?

Excitation Table of SR FF

Output		Inputs	
Q_n	Q_{n+1}	S	R
0	0	0	d
0	1	1	0
1	0	0	1
1	1	d	0

Excitation Table gives all possibilities of present state, next state and the required inputs to have such change from present state to next state.

Reduced Truth Table of JK Flip-Flop

Inputs		Output	Comment
J_n	K_n	Q_{n+1}	
0	0	Q_n	No Change
0	1	0	Reset
1	0	1	Set
1	1	$\overline{Q_n}$	Toggle

Derivation of Excitation Table of JK Flip-Flop

Inputs		Output	
J	K	P/S Q_n	N/S Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

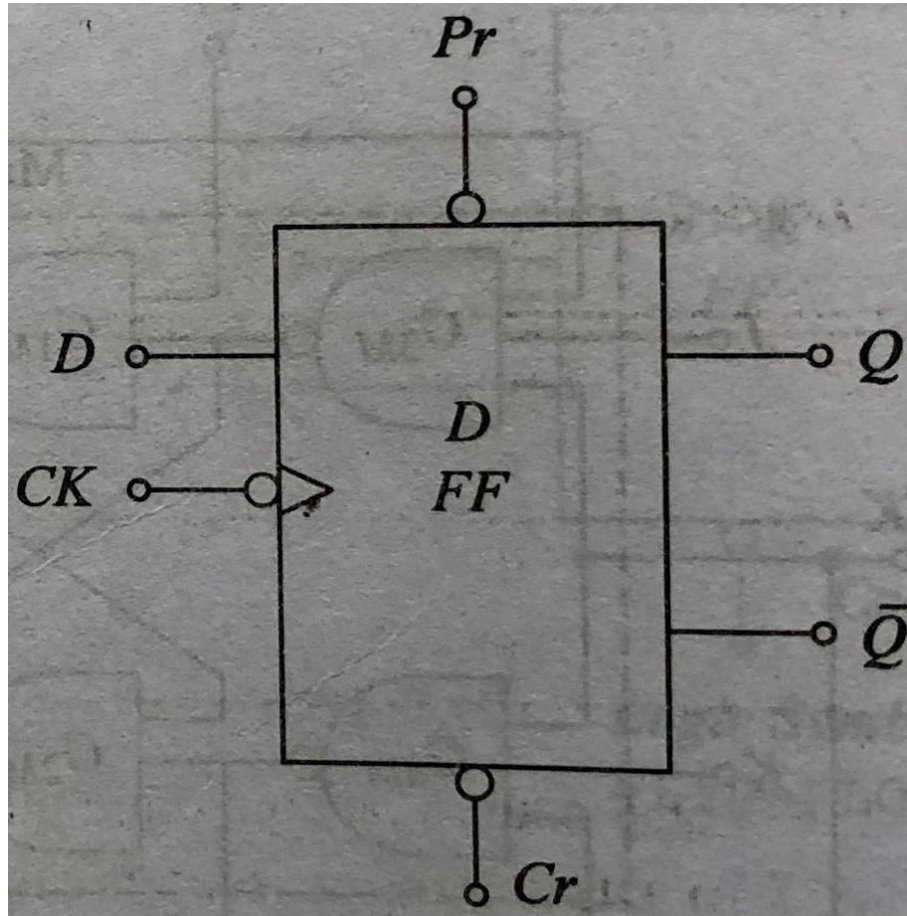
Excitation Table of JK FF

Output		Inputs	
Q_n	Q_{n+1}	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

Excitation Table gives all possibilities of present state, next state and the required inputs to have such change from present state to next state.

Logic symbol of D Flip-Flop

TT of D FF



Input D_n	Output Q_{n+1}
0	0
1	1

In D FF Next State Output is Same as D

Derivation of Excitation Table of D Flip-Flop

Input	Output	
D	P/S Q_n	N/S Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table of D FF

Output		Input
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table gives all possibilities of present state, next state and the required inputs to have such change from present state to next state.

TT of T FF

Input T_n	Output Q_{n+1}
0	Q_n
1	$\overline{Q_n}$

When $T = 1$, For each clock pulse output is complement of previous state. It acts as a Toggle switch, because of this it is called as T (Toggle) FF.

Derivation of Excitation Table of T Flip-Flop

Input	Output	
T	P/S Q_n	N/S Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table of T FF

Output		Input
Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table gives all possibilities of present state, next state and the required inputs to have such change from present state to next state.

Convert SR Flip-Flop to JK Flip-Flop

Required FF Inputs		Outputs		Given FF Inputs	
J	K	Q _n	Q _{n+1}	S	R
0	0	0	0	0	d
0	0	1	1	d	0
0	1	0	0	0	d
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	d	0
1	1	0	1	1	0
1	1	1	0	0	1

Find next state output
using TT of required (JK) FF

Find SR inputs using Ex
table of SR FF

K Map and Logic Diagram

J \ KQ _p	00	01	11	10
	0	1	3	2
0	0	X	0	0
1	1	X	0	1

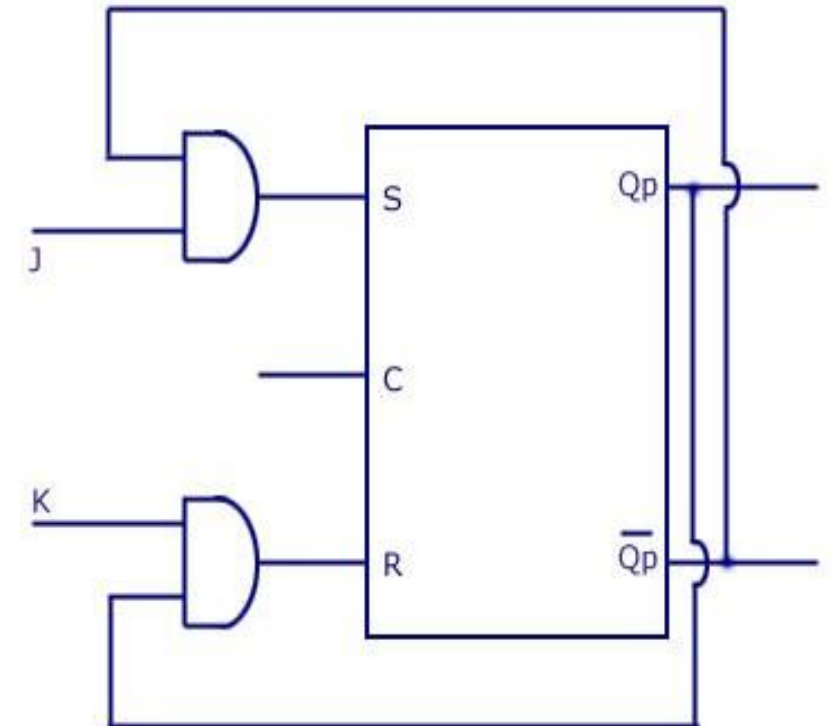
$$S = \bar{J}Q_p$$

J \ KQ _p	00	01	11	10
	0	1	3	2
0	X	0	1	X
1	0	0	1	0

$$R = KQ_p$$

K-Map

Logic Diagram



Convert JK Flip-Flop to T Flip-Flop

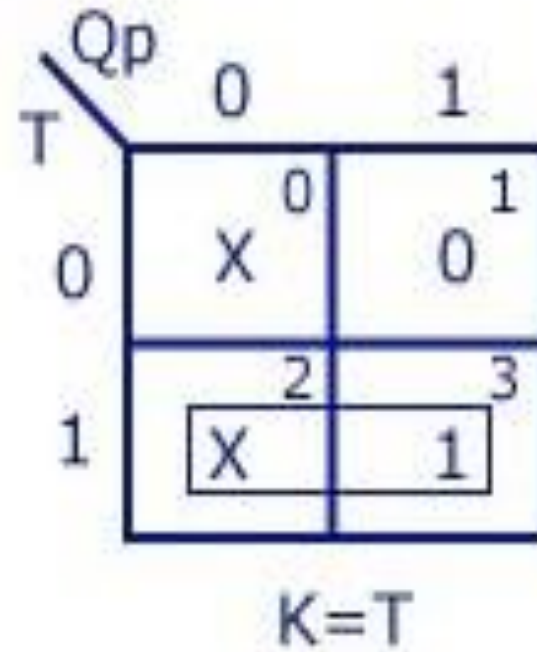
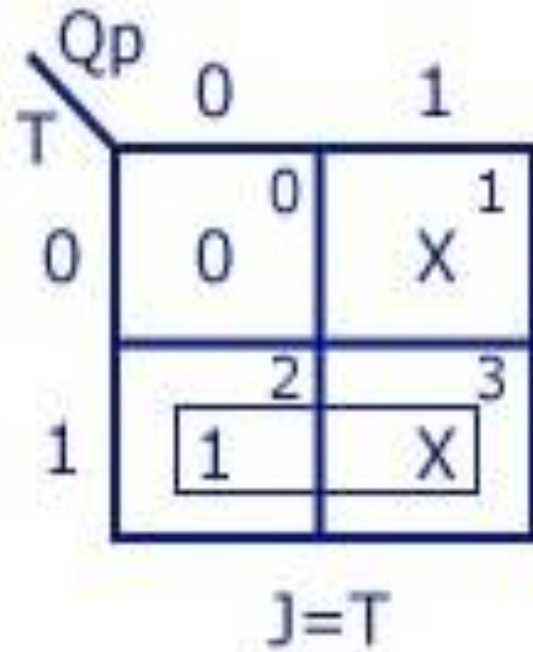
Required FF Input	Outputs		Given FF Inputs	
T	Q _n	Q _{n+1}	J	K
0	0	0	0	d
0	1	1	d	0
1	0	1	1	d
1	1	0	d	1

Find next state output using
TT of required (T) FF

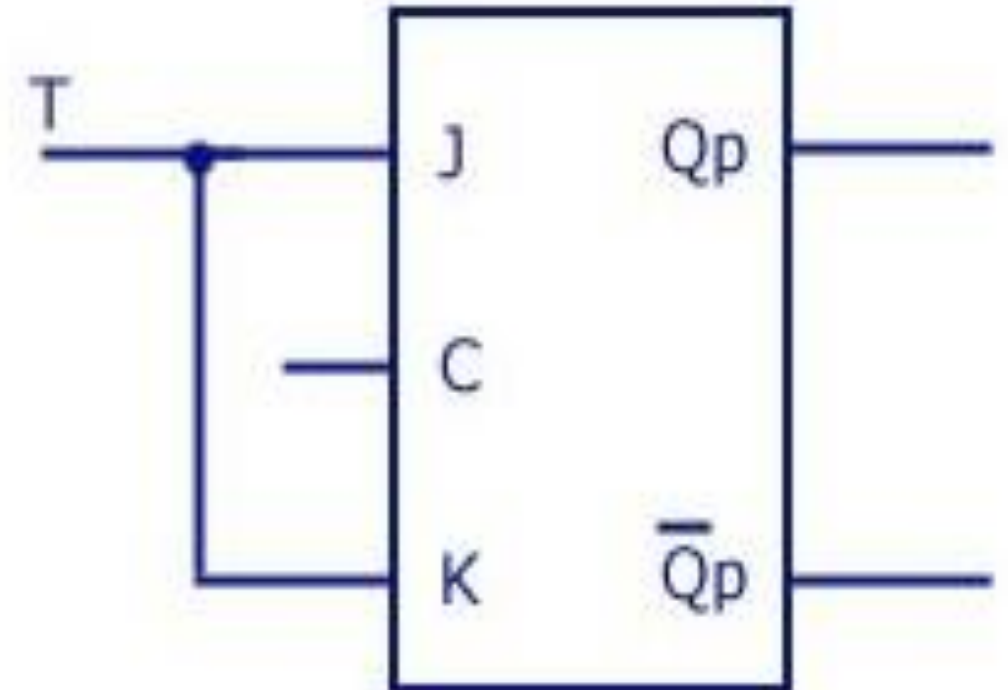
Find JK inputs using Ex
table of JK FF

K Maps and JK FF converted to T FF Diagram

K-maps



Logic Diagram



Convert JK Flip-Flop to D Flip-Flop

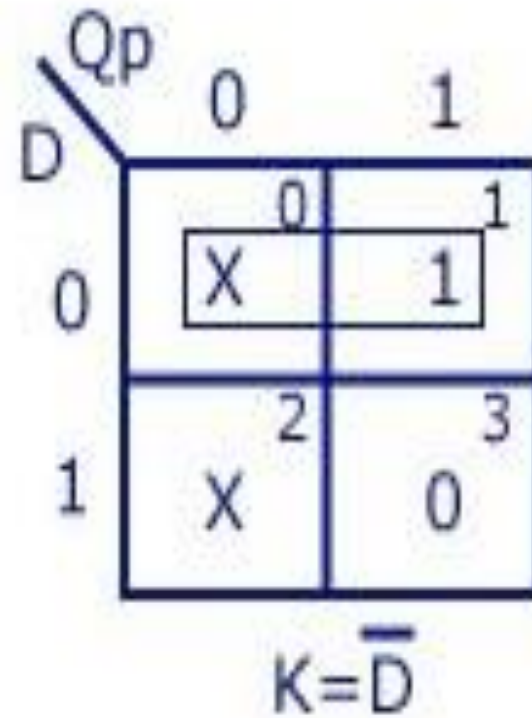
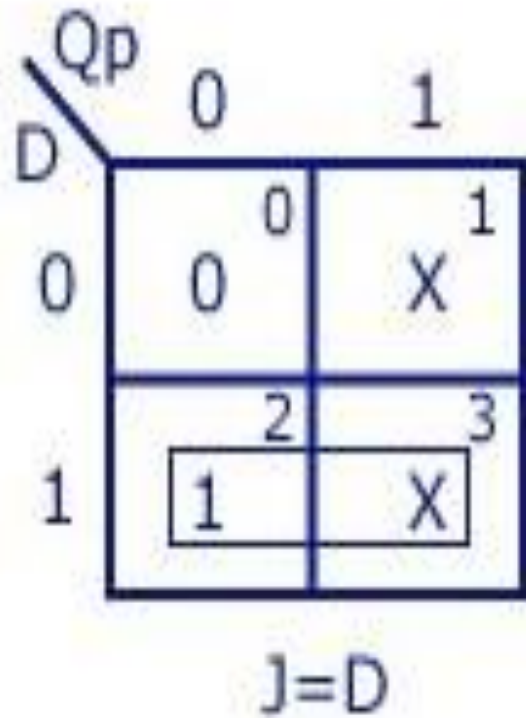
Required FF Input	Outputs		Given FF Inputs	
D	Q _n	Q _{n+1}	J	K
0	0	0	0	d
0	1	0	d	1
1	0	1	1	d
1	1	1	d	0

Find next state output using
TT of required (D) FF

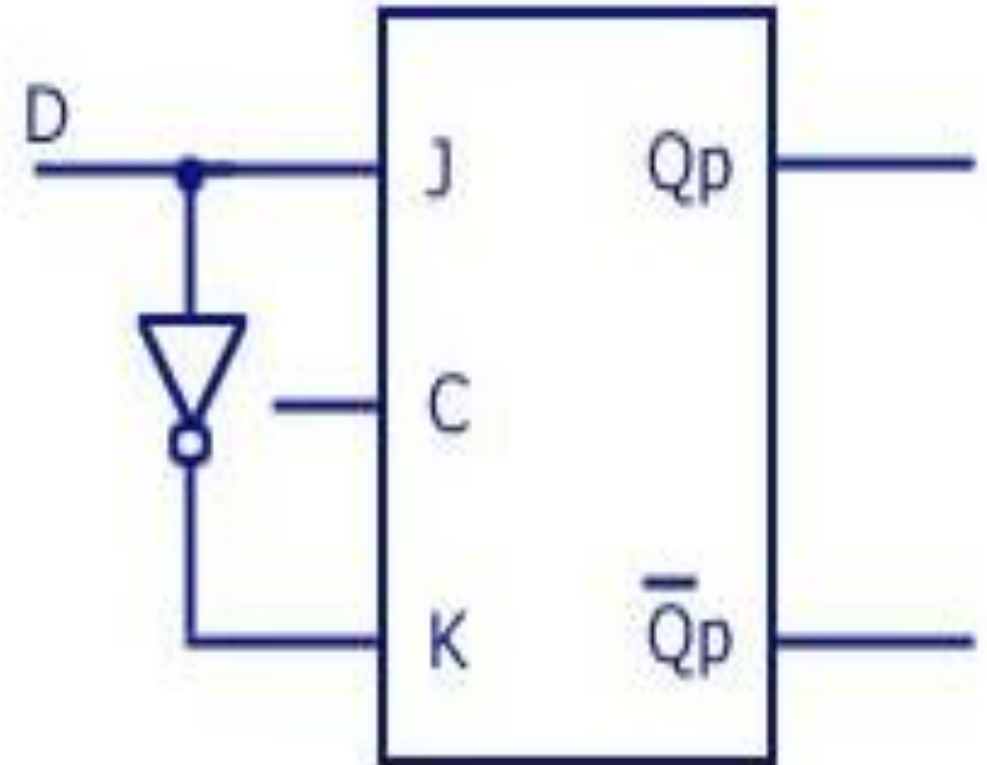
Find JK inputs using Ex
table of JK FF

K Maps and JK FF converted to D FF Diagram

K-maps



Logic Diagram



Convert D Flip-Flop to JK Flip-Flop

Required FF Inputs		Outputs		Given FF Input
J	K	Q _n	Q _{n+1}	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

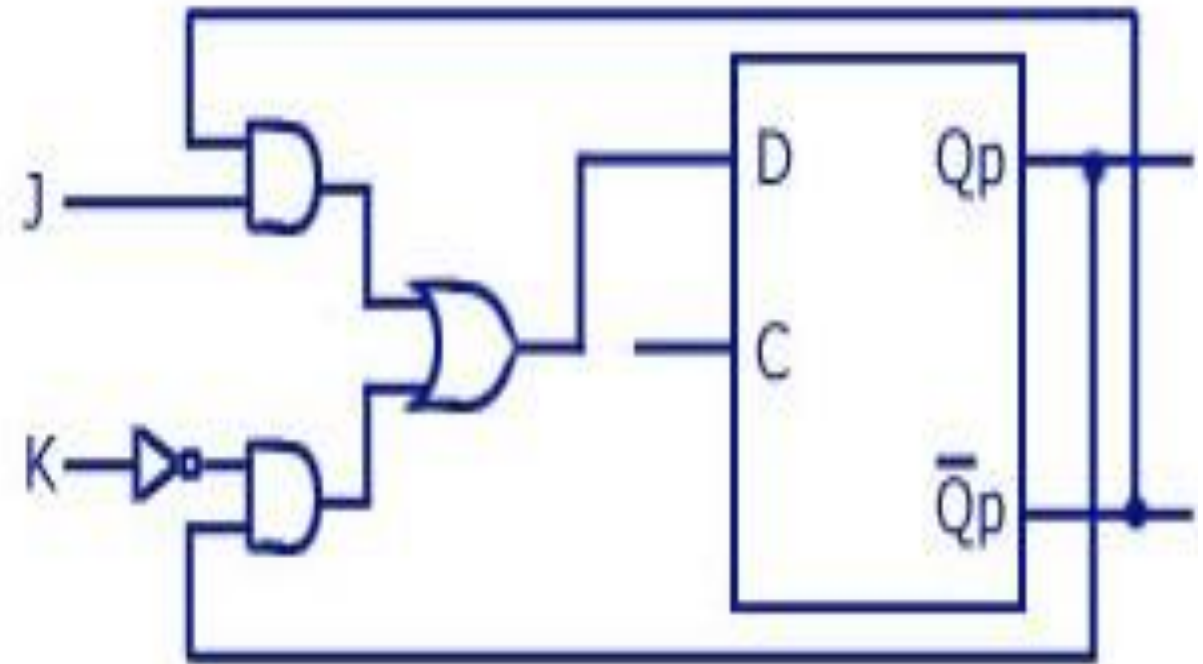
K Map and D FF converted to JK FF Diagram

K-map

KQp					
		00	01	11	10
J	0	0	1	0	0
	1	0	1	0	0
		4	5	7	6
		1	1	0	1

$$D = J\bar{Q}_p + \bar{K}Q_p$$

Logic Diagram



Counters

Counter is a logic circuit used to count number of clock pulses arriving at its clock input pins.

There are two types of counters:

- 1) Synchronous counter
- 2) Asynchronous counter

Synchronous counter:- In this clock is applied simultaneously to all the flip-flops.

Asynchronous counter:- In this clock is **not** applied simultaneously to all the flip-flops. The FF to which clock is applied gives LSB.

Asynchronous counter or Ripple Counter

Design of Mod 4 asynchronous counter

Modulus of counter:- Modulus of counter or counter cycle length means the number of different states the counter has.

How to calculate number of flip-flops required for the counter

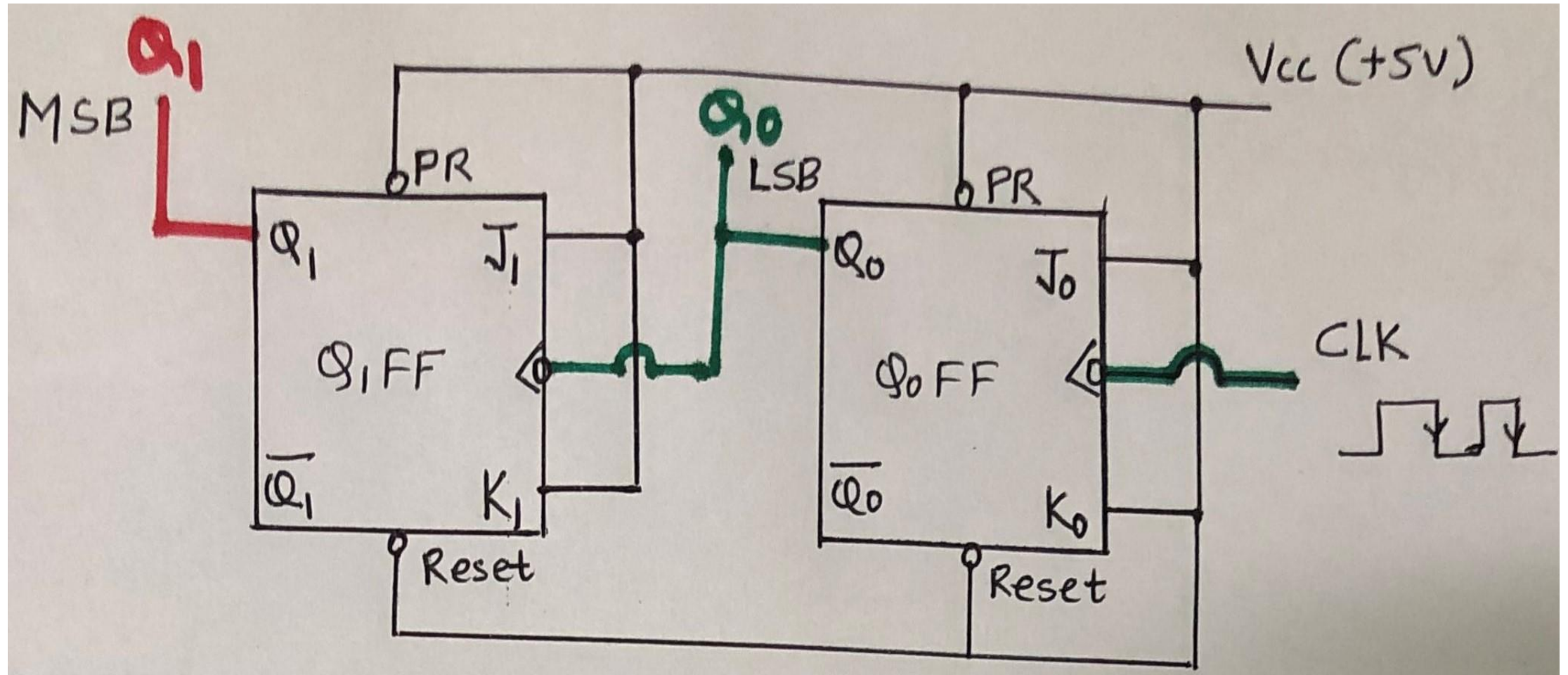
$$2^n \geq N \geq 2^{n-1}$$

Where **n** is number of flip-flops and **N** is counter cycle length.

For mod 4 counter; **N = 4** and the minimum value of **n** for which the above equation is satisfied is 2. Hence 2 flip-flops are required for mod 4 counter.

As clock ripples through the FFs, the Asynchronous counter is also called as Ripple counter

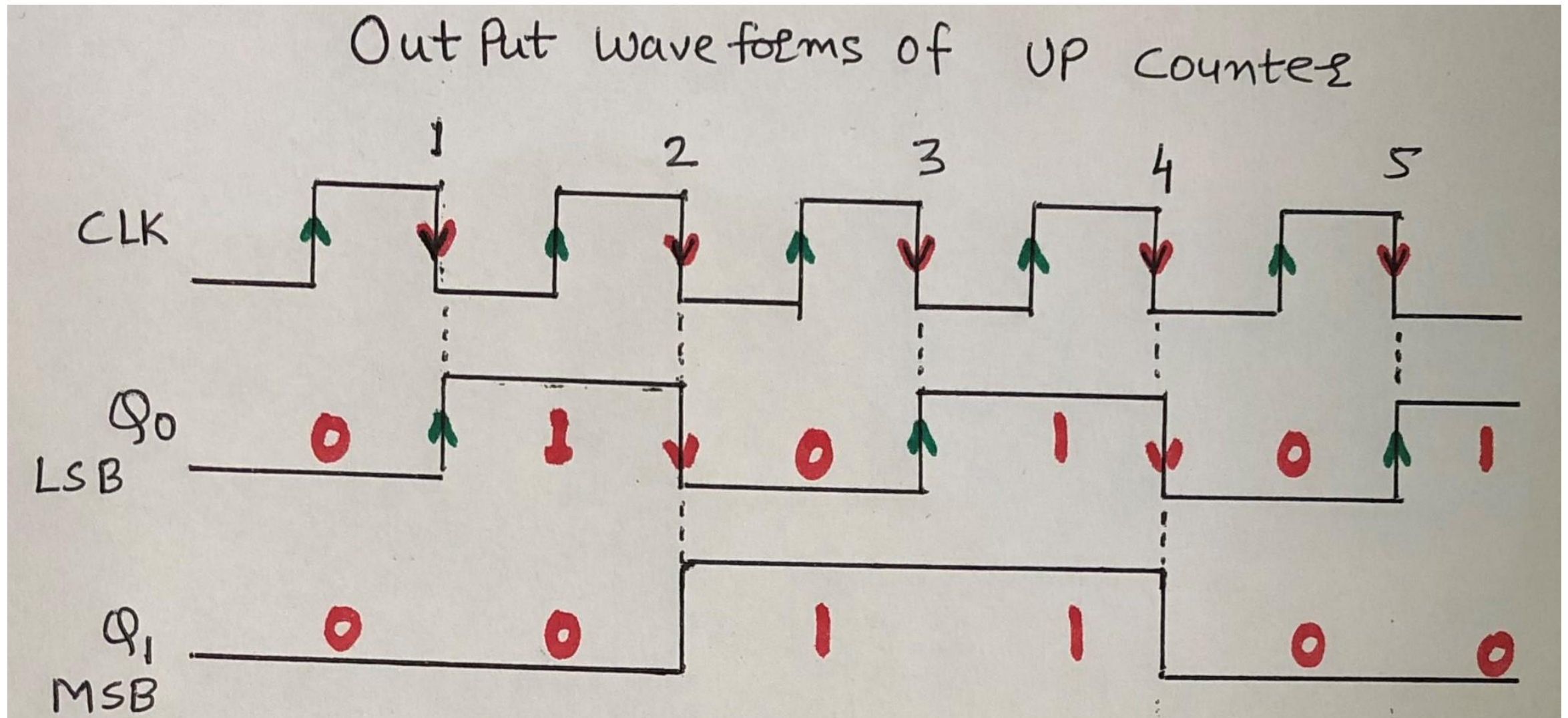
Mod 4 UP Asynchronous (Ripple) counter



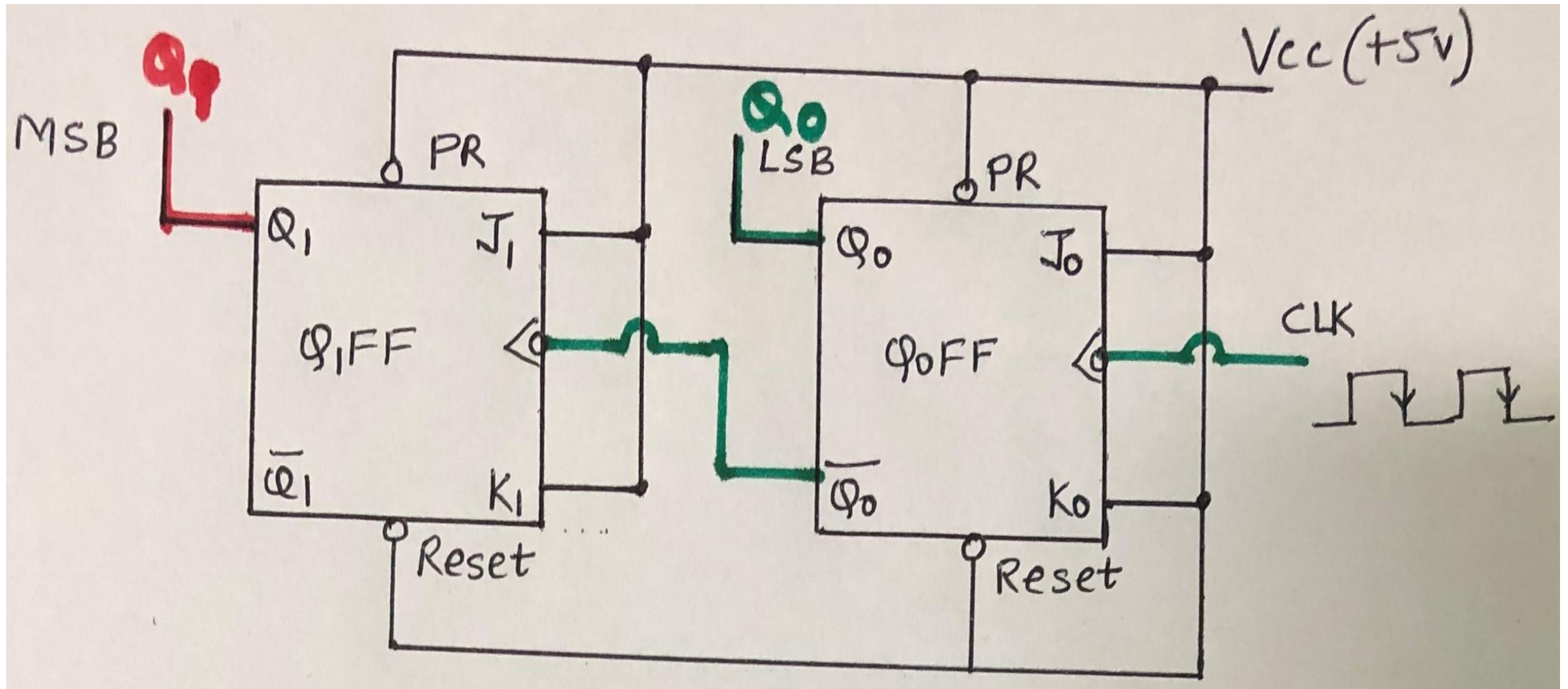
Counting sequence of mod 4 up counter

Clk	Counting Sequence	FF Outputs	
		Q1	Q0
	0	0	0
1	1	0	1
2	2	1	0
3	3	1	1
4	0	0	0

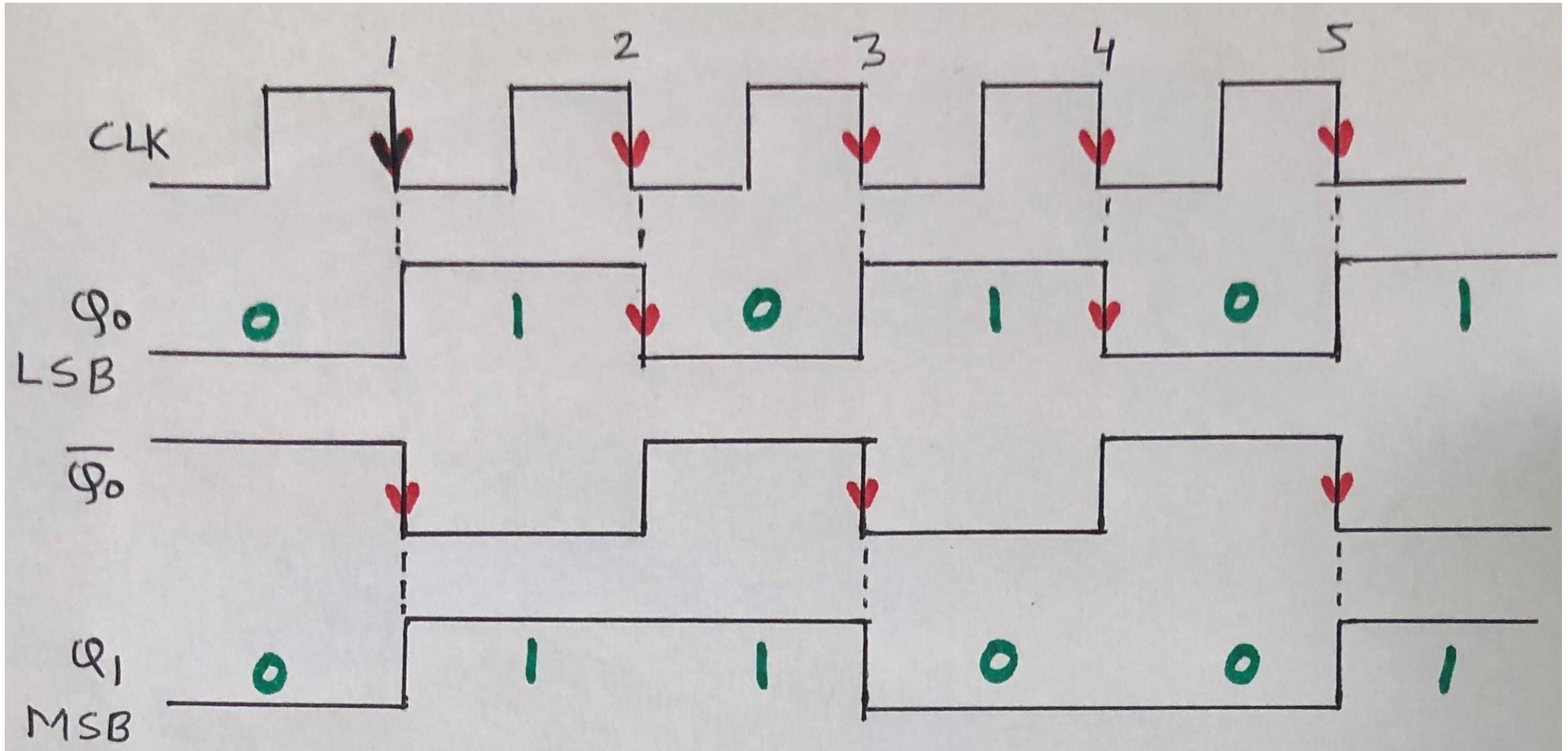
Waveforms of Mod 4 UP asynchronous counter



Mod 4 Down Asynchronous (Ripple) counter



Waveforms of Mod 4 Down asynchronous counter



Counting sequence of mod 4 Down counter

Clk	Counting Sequence	FF Outputs	
		Q1	Q0
	0	0	0
1	3	1	1
2	2	1	0
3	1	0	1
4	0	0	0

Design of Mod 6 Up asynchronous counter

How to calculate number of flip-flops required for the Mod 6 counter

$$2^n \geq N \geq 2^{n-1}$$

Where n is number of flip-flops and N is counter cycle length.

For mod 6 counter; $N = 6$ and the minimum value of n for which the above equation is satisfied is 3. Hence 3 flip-flops are required for mod 6 counter.

Method to break the natural counting sequence

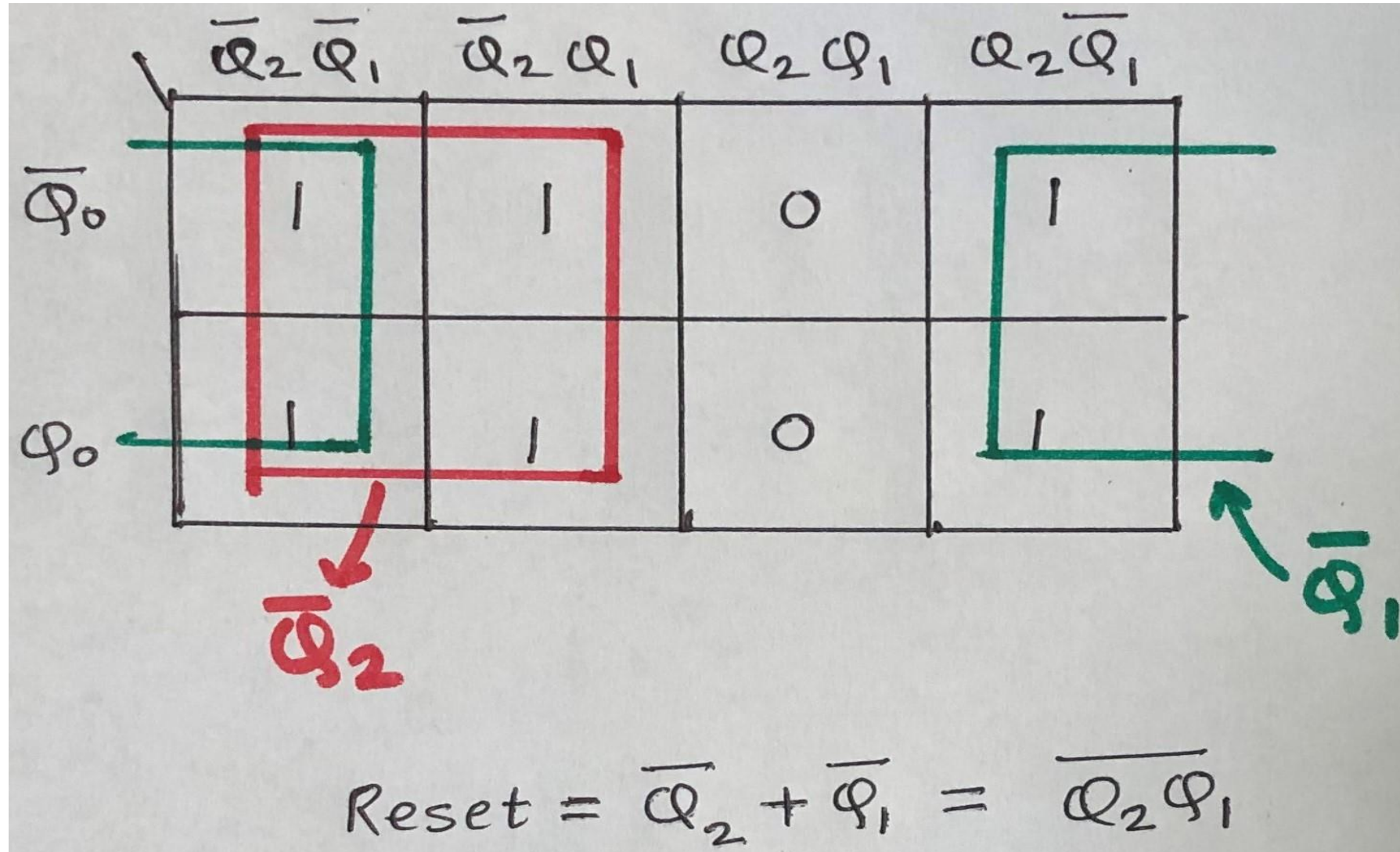
Counting Sequence	FF Outputs			Reset Signal
	Q2	Q1	Q0	
0	0	0	0	1
1	0	0	1	1
2	0	1	0	1
3	0	1	1	1
4	1	0	0	1
5	1	0	1	1
6	1	1	0	0
7	1	1	1	0

counting sequence of Mod 6 UP counter

Clk	FF Outputs			Counting Sequence
	Q2	Q1	Q0	
	0	0	0	0
1	0	0	1	1
2	0	1	0	2
3	0	1	1	3
4	1	0	0	4
5	1	0	1	5
6	0	0	0	0

Highest number for Mod 6 Up counter is 5

K map to design circuit for reset pins



Power-ON Reset, Preset circuit

Power-ON Reset circuit to start the counter from initial state.

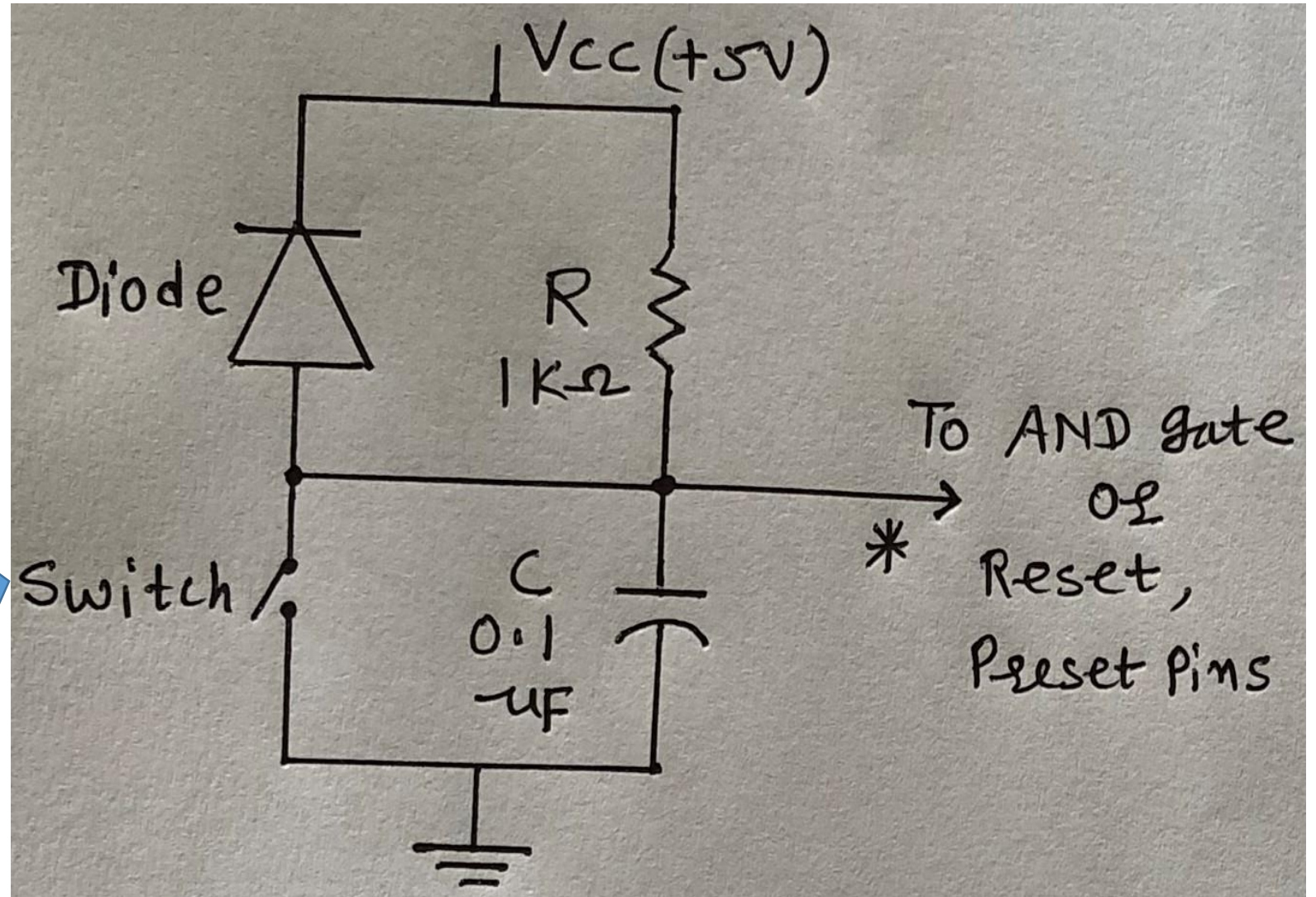
To start counter from 000, use the following Power-ON reset ckt. connect o/p of AND gate to Reset pins of all the FFs.

To start counter from 111, connect o/p of AND gate to Preset pins of all FFs.

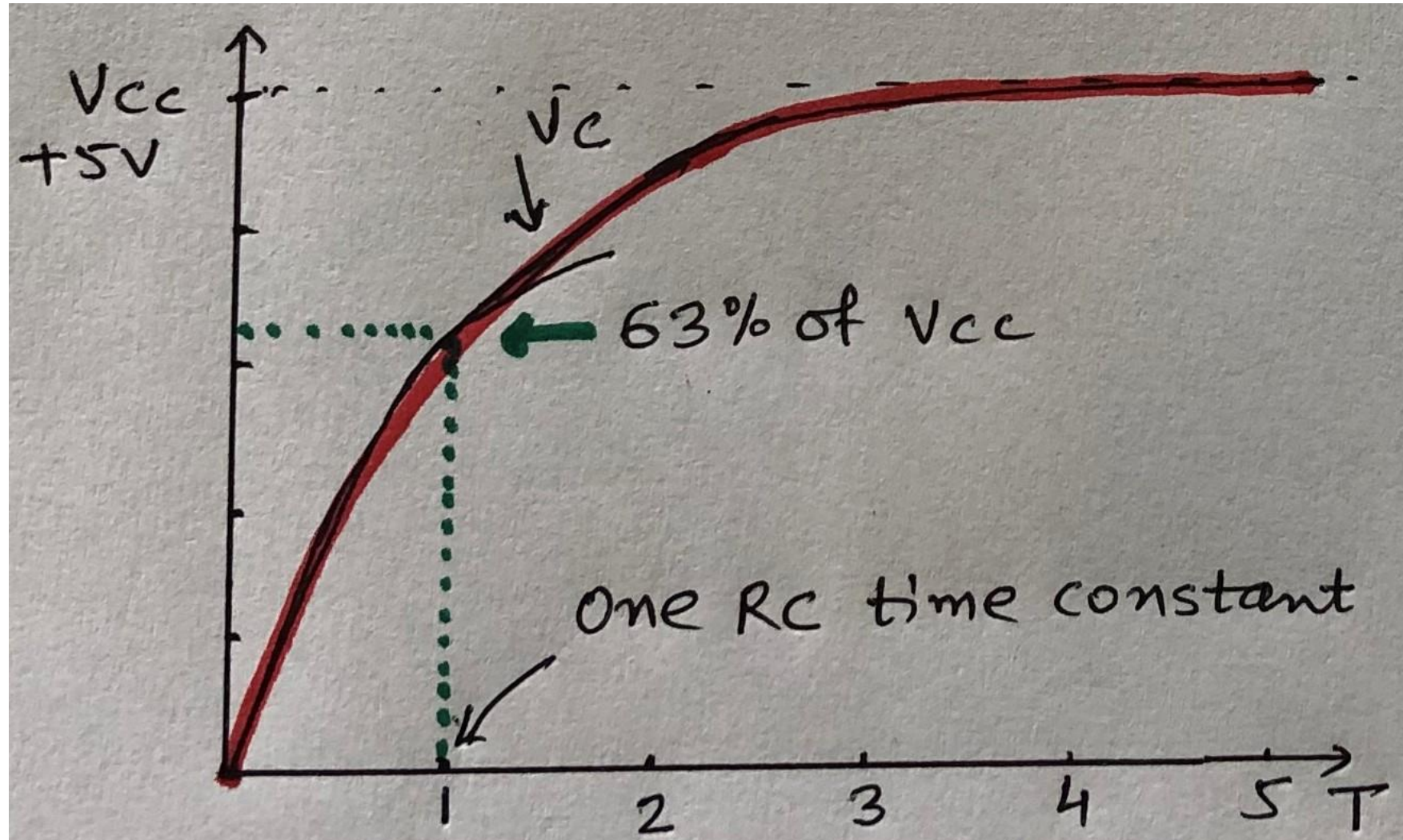
If all the states are used then connect o/p of Power-ON reset ckt to Reset pins of all FFs directly.

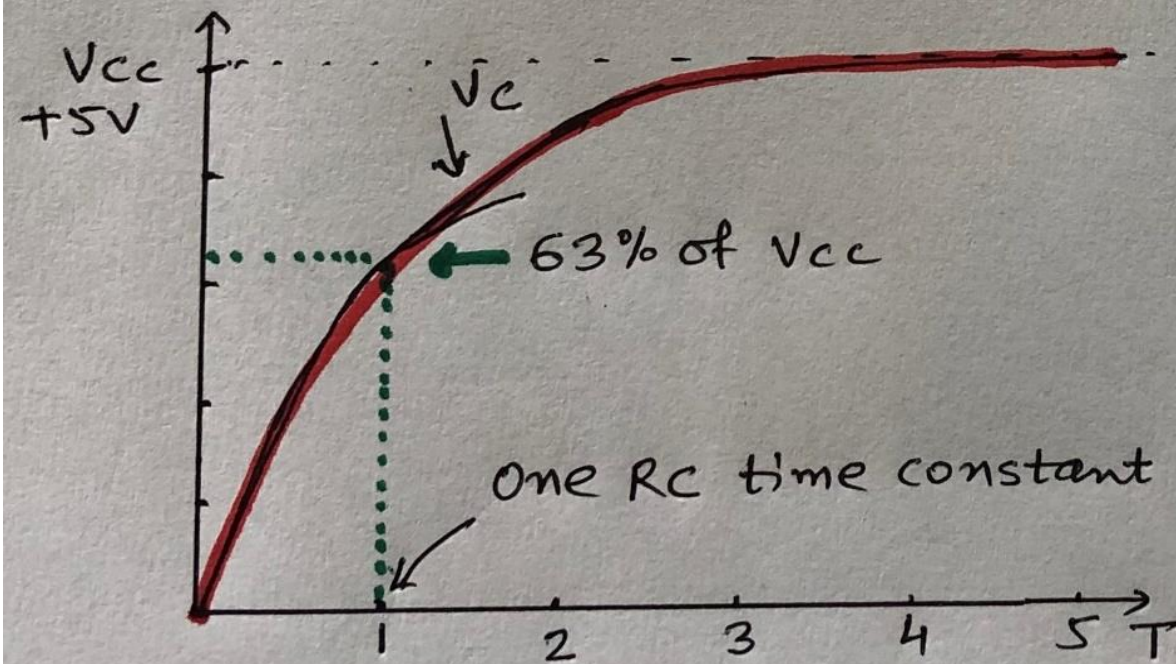
Power-ON Reset, Preset circuit

When switch is pressed the capacitor will discharge and low voltage is applied to reset pins. This helps in resetting the counter any time



Capacitor charging waveform of Power-ON Reset, Preset circuit

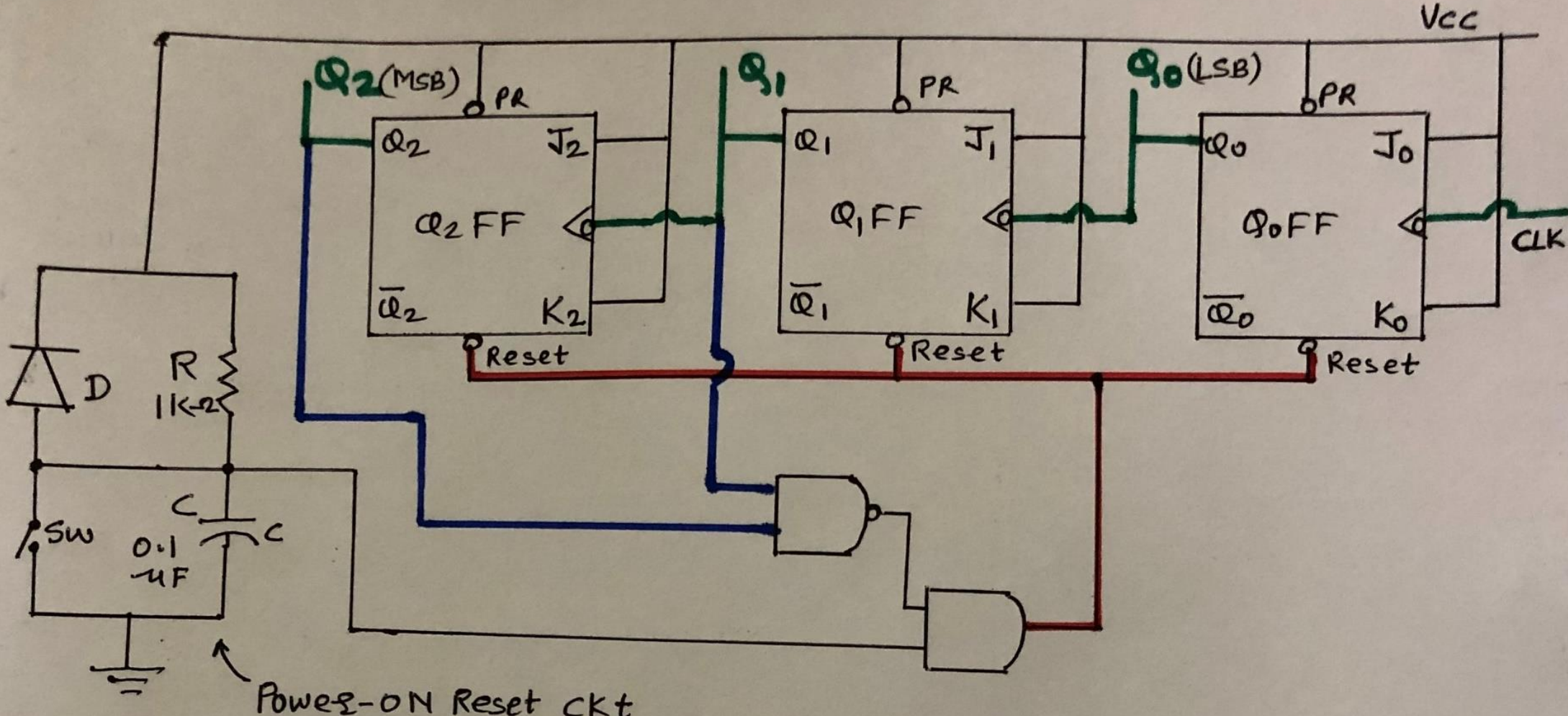




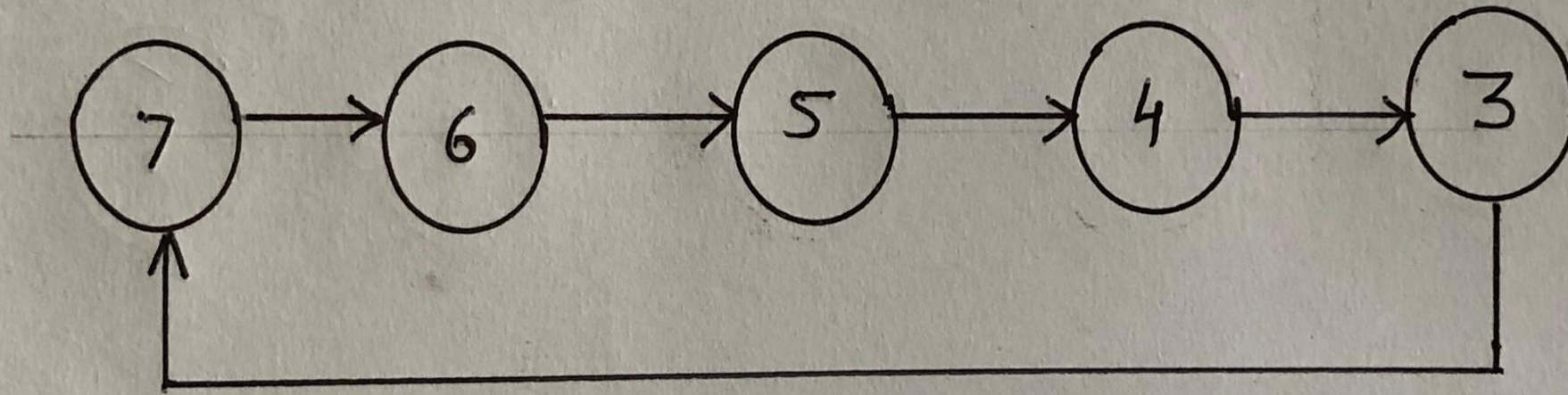
Initially when power supply is turned ON, the voltage across the capacitor C will be zero. This will reset the FF if $*$ is connected to reset pins and preset the FF if $*$ is connected to preset pins.

After 1 RC time constant, the voltage V_C will be $\approx 3V$, which is recognized as high level signal. This will disable the reset, preset pins, and normal operation of FF will take place. [Don't connect $*$ to both Reset, preset pins of same FF]

Mod 6 UP asynchronous counter with power-on Reset facility.

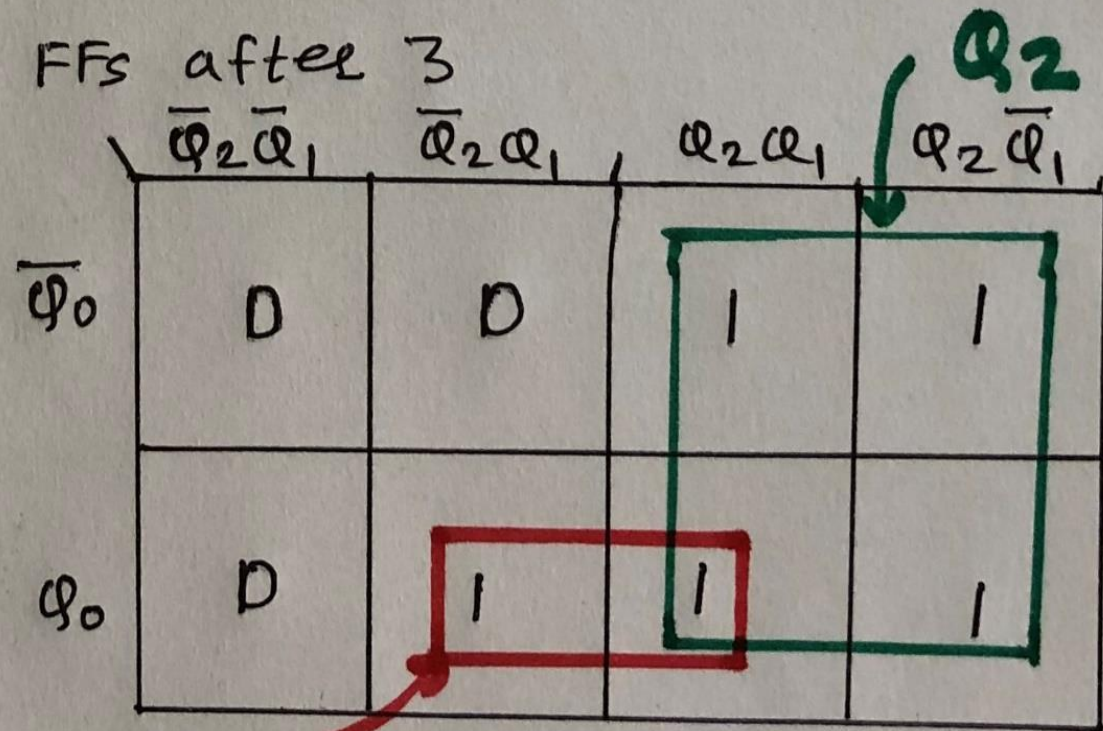


Design Ripple counter (Asynchronous counter) using JKMS
FFs for the following state diagram;

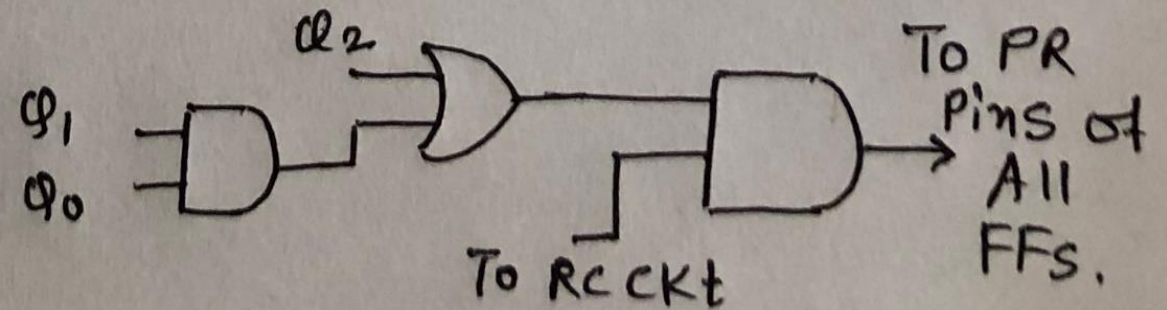


Ckt design to Preest all FFs after 3

Q_2	Q_1	Q_0	Preset
1	1	1	1
1	1	0	1
1	0	1	1
1	0	0	1
0	1	1	1
0	1	0	0
0	0	1	0
0	0	0	0

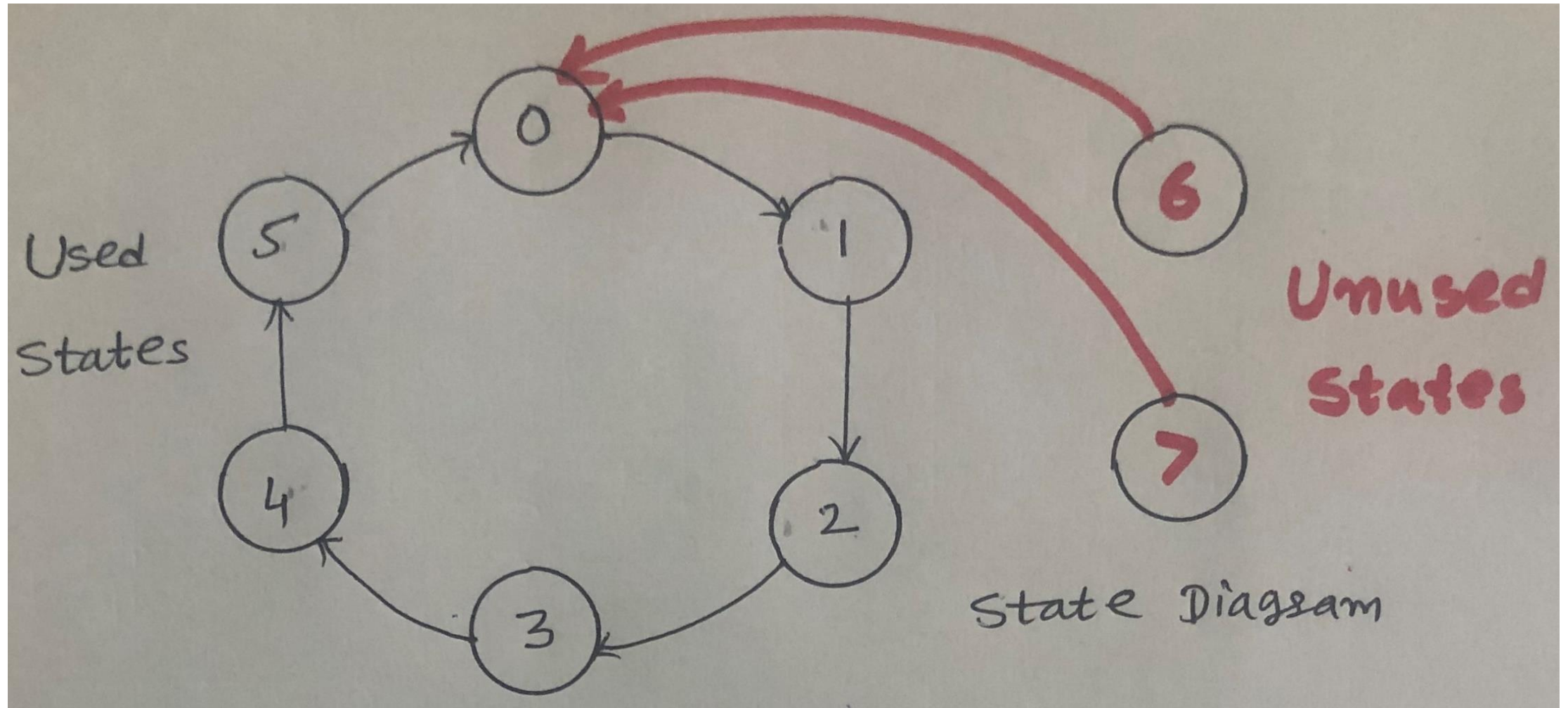


Q_1, Q_0



Synchronous Counter Design

Design Mod 6 Up Synchronous Counter using JKMS Flip-Flops.



Mod 6 Up Synchronous Counter using JKMS FFs

$$2^n \geq N \geq 2^{n-1}$$

3 FFs are required

Present State			Next State			Flip-Flop Inputs					
A	B	C	A	B	C	J _A	K _A	J _B	K _B	J _C	K _C
0	0	0	0	0	1	0	d	0	d	1	d
0	0	1	0	1	0	0	d	1	d	d	1
0	1	0	0	1	1	0	d	d	0	1	d
0	1	1	1	0	0	1	d	d	1	d	1
1	0	0	1	0	1	d	0	0	d	1	d
1	0	1	0	0	0	d	1	0	d	d	1
1	1	0	0	0	0	d	1	d	1	0	d
1	1	1	0	0	0	d	1	d	1	d	1

K Maps

	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}$	0	0	d	d
C	0	1	d	d

$J_A = BC$

	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}$	d	d	1	0
C	d	d	1	1

$K_A = B + C$

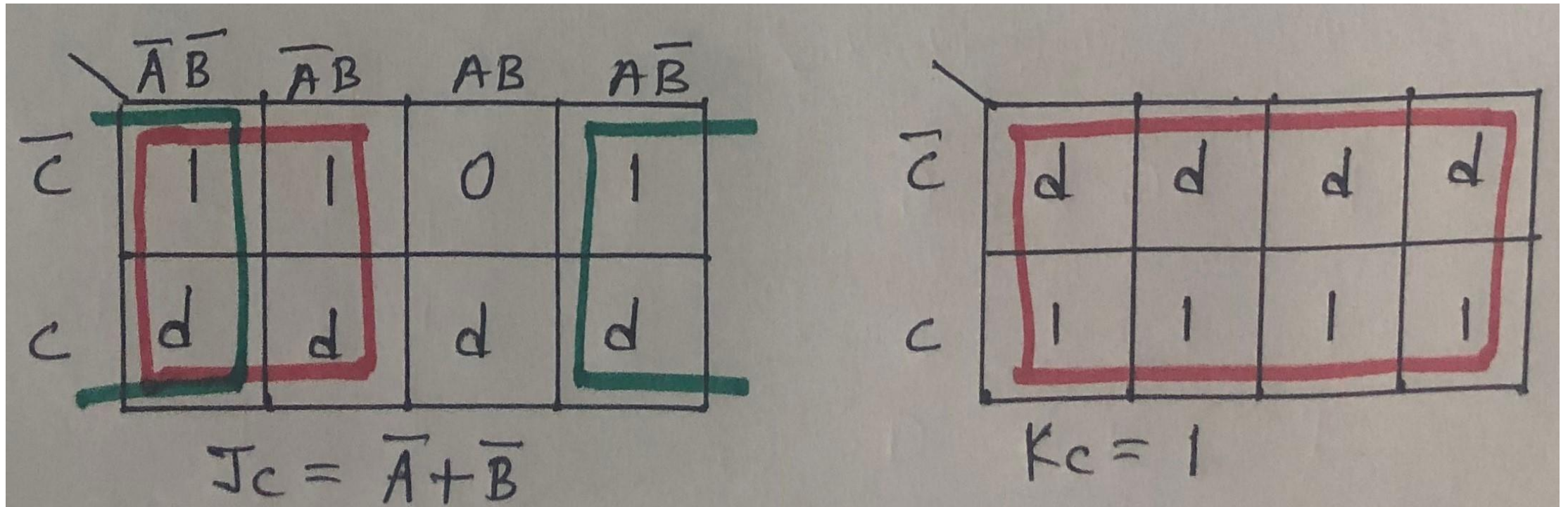
	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}$	0	d	d	0
C	1	d	d	0

$J_B = \bar{A}C$

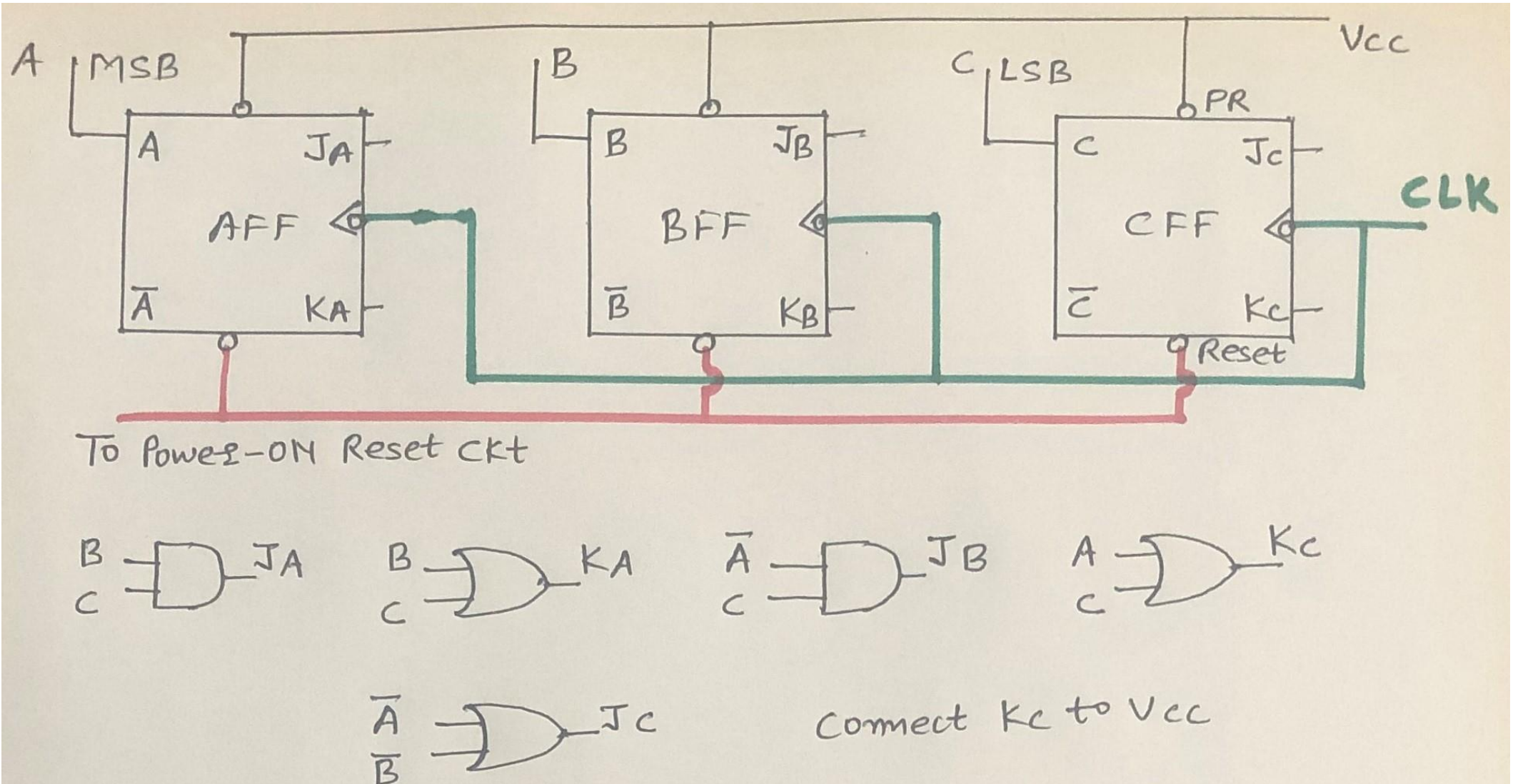
	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}$	d	0	1	d
C	d	1	1	d

$K_B = A + C$

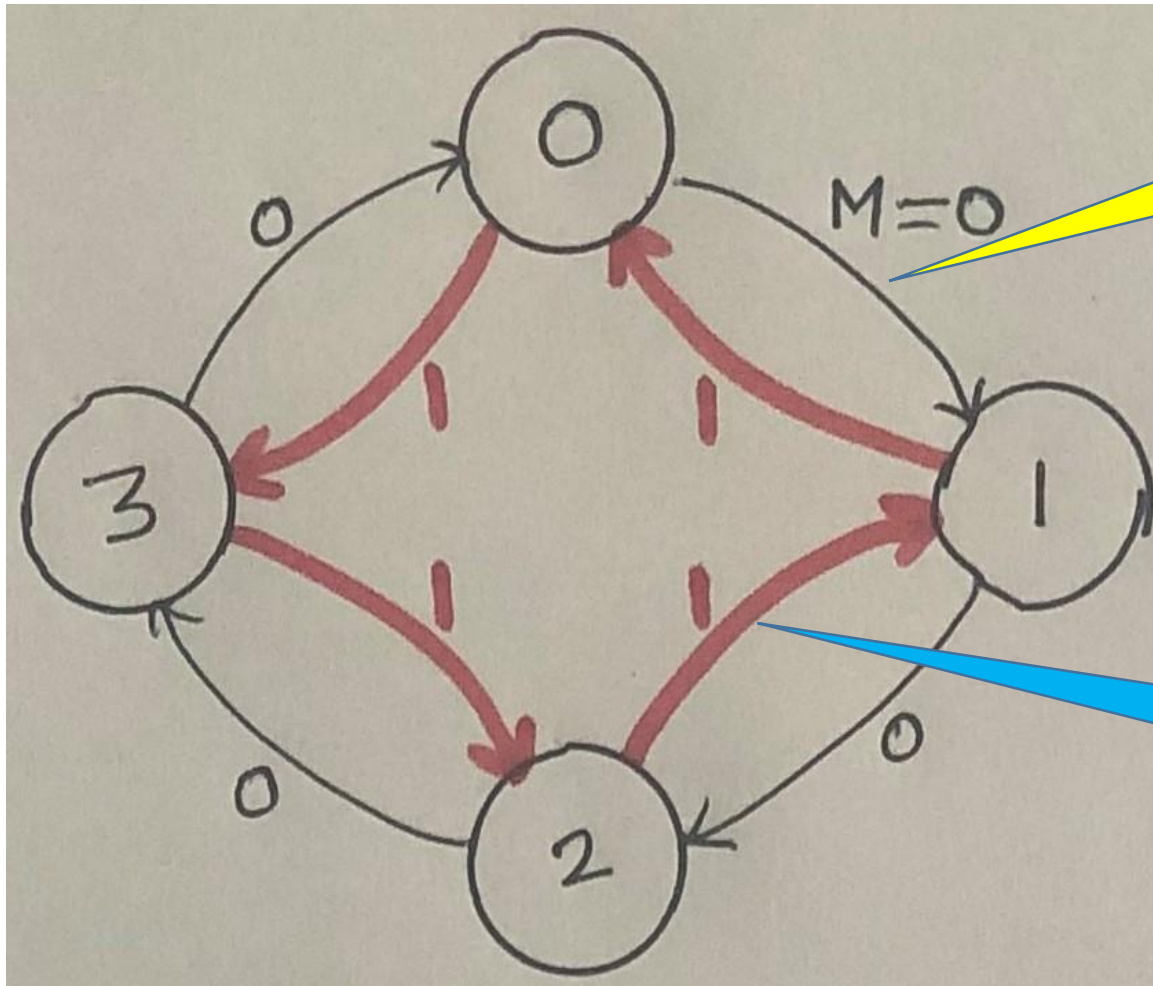
K Maps



Logic Diagram of Mod 6 Up Lockout Free Synchronous Counter



Design Mod 4 Mode Select Up/Down Synchronous Counter using JKMS Flip-Flops.



M = 0, the counter should go in Up direction

M = 1, the counter should go in Down direction

Mod 4 Mode select synchronous counter

$$2^n \geq N \geq 2^{n-1}$$

2 FFs are required

Mode Control	Present State		Next State		Flip-Flop Inputs			
M	A	B	A	B	JA	KA	JB	KB
0	0	0	0	1	0	d	1	d
0	0	1	1	0	1	d	d	1
0	1	0	1	1	d	0	1	d
0	1	1	0	0	d	1	d	1
1	0	0	1	1	1	d	1	d
1	0	1	0	0	0	d	d	1
1	1	0	0	1	d	1	1	d
1	1	1	1	0	d	0	d	1

K Maps

	$\bar{M}\bar{A}$	$\bar{M}A$	MA	$M\bar{A}$
$\bar{B}$	0	d	d	1
B	1	d	d	0

$$J_A = \bar{M}B + M\bar{B}$$

$$J_A = M \oplus B$$

1	1	1	1
d	d	d	d

$$J_B = 1$$

	$\bar{M}\bar{A}$	$\bar{M}A$	MA	$M\bar{A}$
$\bar{B}$	d	0	1	d
B	d	1	0	d

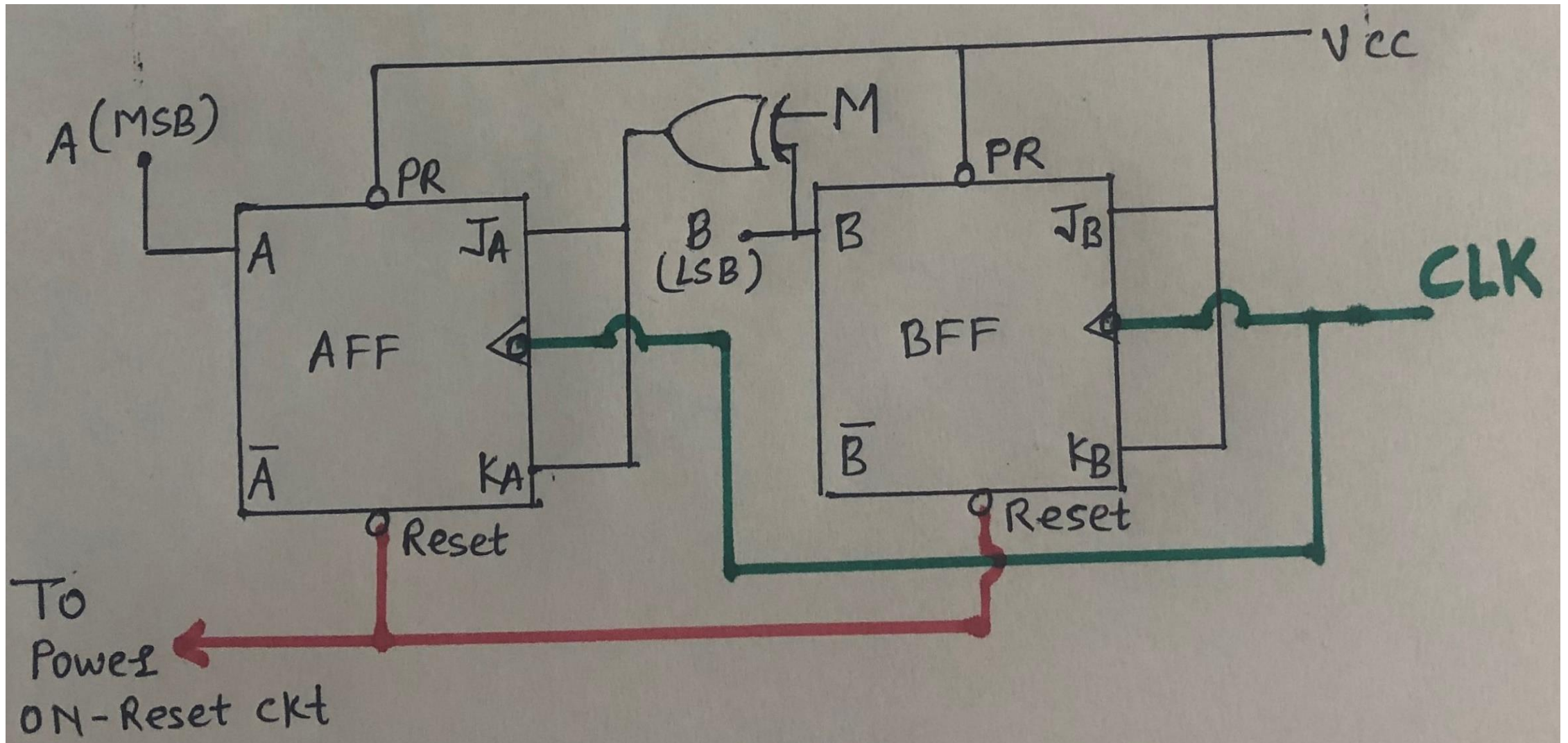
$$K_A = \bar{M}B + M\bar{B}$$

$$K_A = M \oplus B$$

d	d	d	d
1	1	1	1

$$K_B = 1$$

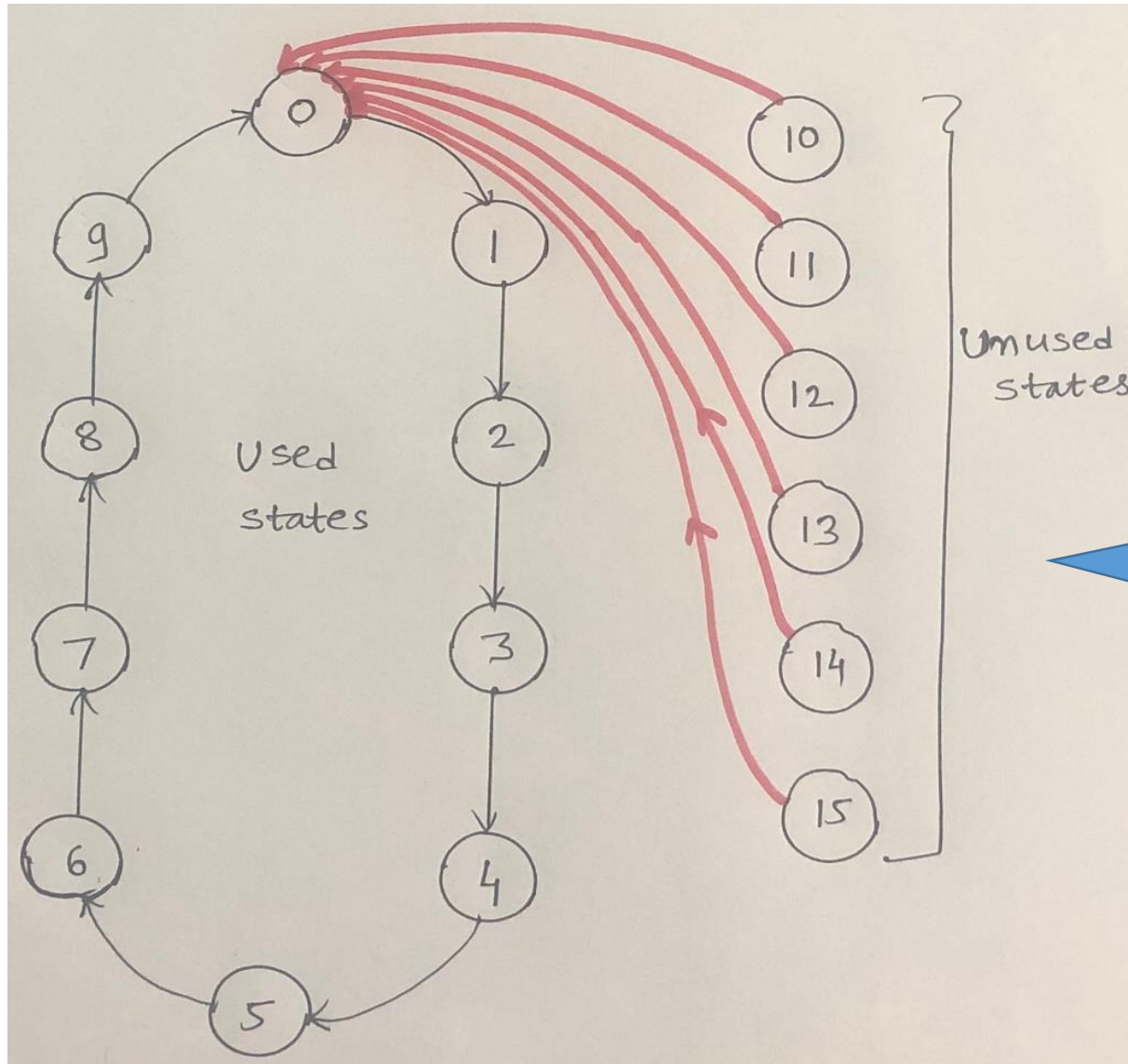
Logic Diagram of Mod 4 Mode Select Up/Down Synchronous Counter



Lockout in Synchronous Counter

- Lockout in counter means if the counter goes to unused state and it remains in unused states without coming back to used states.
- The lockout can be avoided by considering used state as next state after each unused state.
- Lockout is present in synchronous counter. There is no lockout problem in asynchronous counter.

Design Mod 10 (Decade) Up Lockout-Free Synchronous Counter.



**State Diagram of Mod 10
(Decade) Up Lockout-Free
Synchronous Counter.**

Present State				Next State				Flip-Flop Inputs							
A	B	C	D	A	B	C	D	JA	KA	JB	KB	JC	KC	JD	KD
0	0	0	0	0	0	0	1	0	d	0	d	0	d	1	d
0	0	0	1	0	0	1	0	0	d	0	d	1	d	d	1
0	0	1	0	0	0	1	1	0	d	0	d	d	0	1	d
0	0	1	1	0	1	0	0	0	d	1	d	d	1	d	1
0	1	0	0	0	1	0	1	0	d	d	0	0	d	1	d
0	1	0	1	0	1	1	0	0	d	d	0	1	d	d	1
0	1	1	0	0	1	1	1	0	d	d	0	d	0	1	d
0	1	1	1	1	0	0	0	1	d	d	1	d	1	d	1
1	0	0	0	1	0	0	1	d	1	0	d	0	d	1	d
1	0	0	1	0	0	0	0	d	1	0	d	0	d	d	1
1	0	1	0	0	0	0	0	d	1	0	d	d	1	0	d
1	0	1	1	0	0	0	0	d	1	0	d	d	1	d	1
1	1	0	0	0	0	0	0	d	1	d	1	0	d	0	d
1	1	0	1	0	0	0	0	d	1	d	1	0	d	d	1
1	1	1	0	0	0	0	0	d	1	d	1	d	1	0	d
1	1	1	1	0	0	0	0	d	1	d	1	d	1	d	1

	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}\bar{D}$	0	0	d	d
$\bar{C}D$	0	0	d	d
CD	0	1	d	d
$C\bar{D}$	0	0	d	d

$$J_A = BCD$$

1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

$$K_A = 1$$

	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}\bar{D}$	0	d	d	0
$\bar{C}D$	0	d	d	0
CD	1	d	d	0
$C\bar{D}$	0	d	d	0

$$J_B = \bar{A}CD$$

d	0	1	d
d	0	1	d
d	1	1	d
d	0	1	d

$$K_B = A + CD$$

	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}\bar{D}$	0	0	0	0
$\bar{C}D$	1	1	0	0
CD	d	d	d	d
$C\bar{D}$	d	d	d	d

$J_c = \bar{A}D$

d	d	d	d
d	d	d	d
1	1	1	1
0	0	1	1

$K_c = A + D$

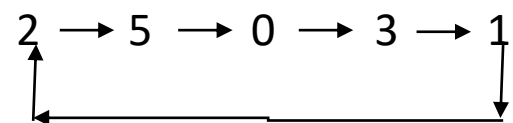
	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}\bar{D}$	1	1	0	1
$\bar{C}D$	d	d	d	d
CD	d	d	d	d
$C\bar{D}$	1	1	0	0

$J_D = \bar{A} + \bar{B}\bar{C}$

d	d	d	d
1	1	1	1
1	1	1	1
d	d	d	d

$K_D = 1$

Design a circuit using JKMS FFs which will generate the following sequence



Present State			Next State			Flip-Flop Inputs					
A	B	C	A	B	C	JA	KA	JB	KB	JC	KC
0	0	0	0	1	1						
0	0	1	0	1	0						
0	1	0	1	0	1						
0	1	1	0	0	1						
1	0	0	0	0	0						
1	0	1	0	0	0						
1	1	0	0	0	0						
1	1	1	0	0	0						

Ring Counter

Ring counter → If the serial o/p Q of the shift register is connected to serial i/p of next FF and o/p of MSB FF is connected to i/p of LSB FF then the injected pulse will keep circulating. This ckt is referred as a ring counter. In case of 4 bit ring counter, 0001 is entered by presetting LSB FF and clearing other 3 FFs. When clk pulses are applied, this 1 circulates around the ckt. so the o/p after each clk pulse will be

0001 → 0010 → 0100 → 1000

↑—————↓

The 4 bit ring counter has only 4 different states
No of states = No of FFs.

No of States = No of FFs.

Design → For designing 4 bit ring counter, 4 FFs are required. D FFs or JK converted D or SR converted to D are required.

P/s				N/s				FF i/p's				Ex table of D FF		
A	B	C	D	A	B	C	D	D _A	D _B	D _C	D _D	p/s	N/s	D
0	0	0	1	0	0	1	0	0	0	1	0	0	0	0
0	0	1	0	0	1	0	0	0	1	0	0	0	1	1
0	1	0	0	1	0	0	0	1	0	0	0	1	0	0
1	0	0	0	0	0	0	1	0	0	0	1	1	1	1

	$\bar{c}\bar{d}$	$\bar{c}d$	cd	$c\bar{d}$
$\bar{A}\bar{B}$	d	0	d	0
$\bar{A}B$	1	d	d	d
AB	d	d	d	d
$A\bar{B}$	0	d	d	d

$$D_A = B$$

d	0	d	1
0	d	d	d
d	d	d	d
0	d	d	d

$$D_B = C$$

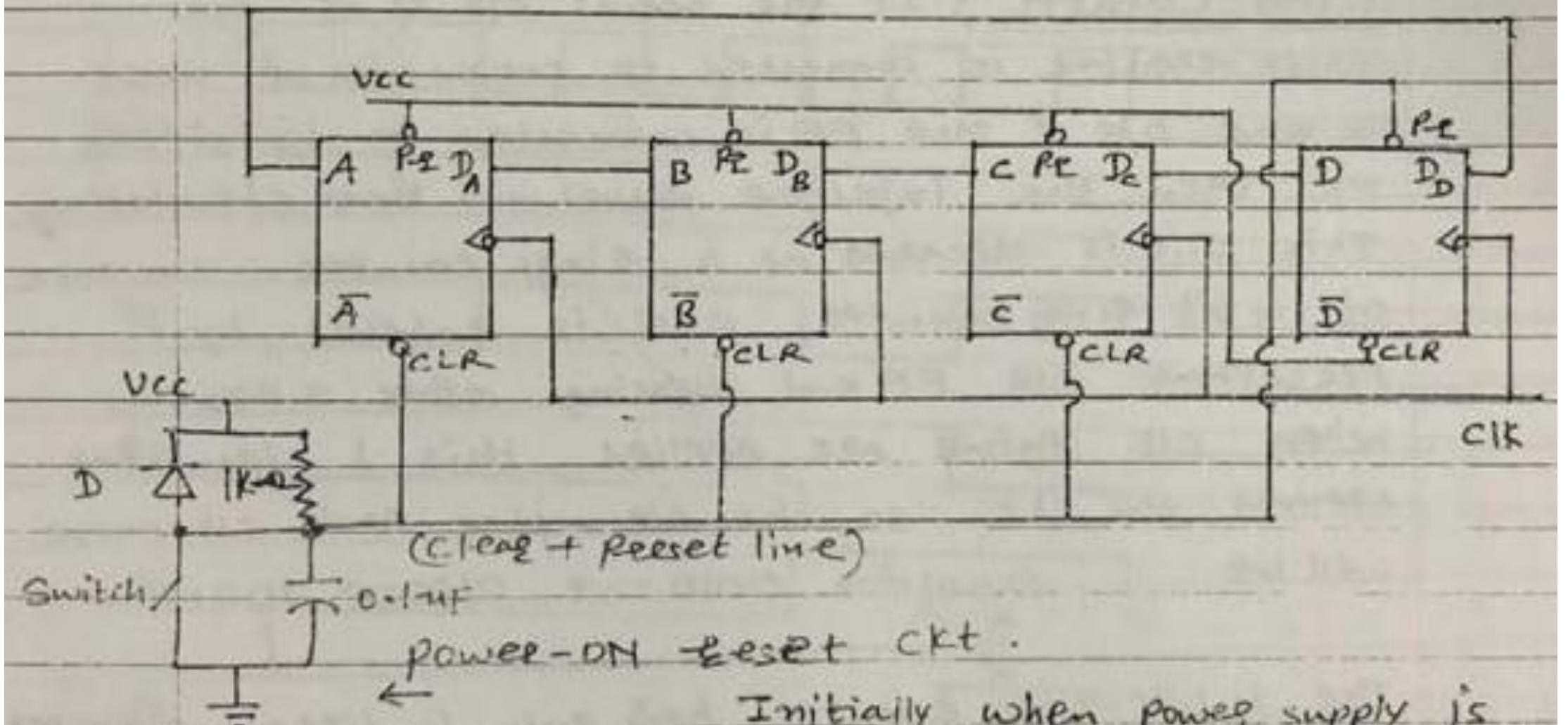
d	1	d	0
0	d	d	d
d	d	d	d
0	d	d	d

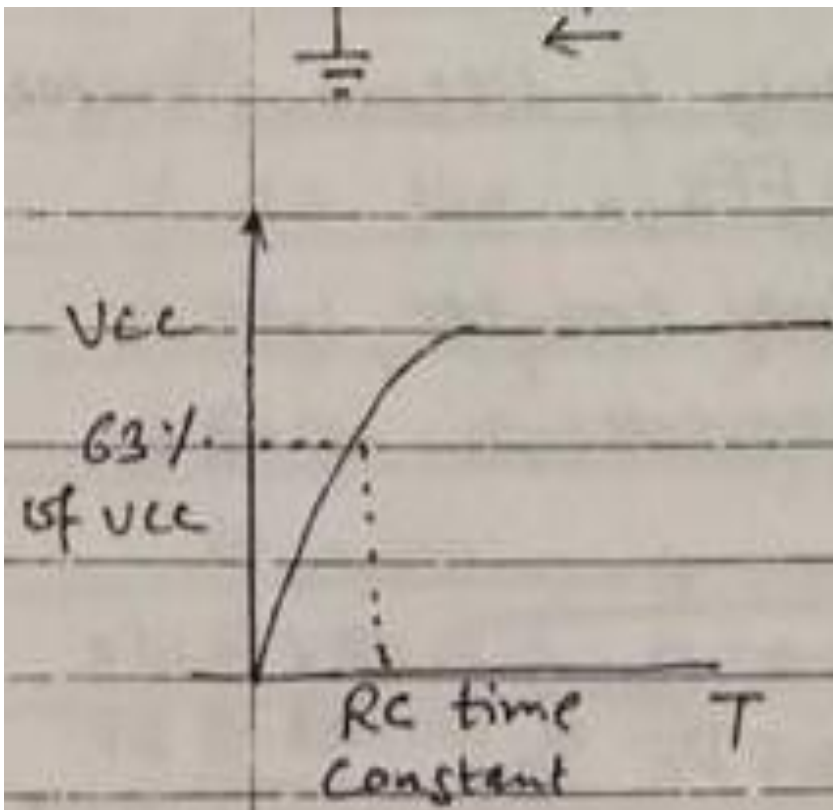
$$D_C = D$$

d	0	d	0
0	d	d	d
d	d	d	d
1	d	d	d

$$D_D = A$$

Logic dia →

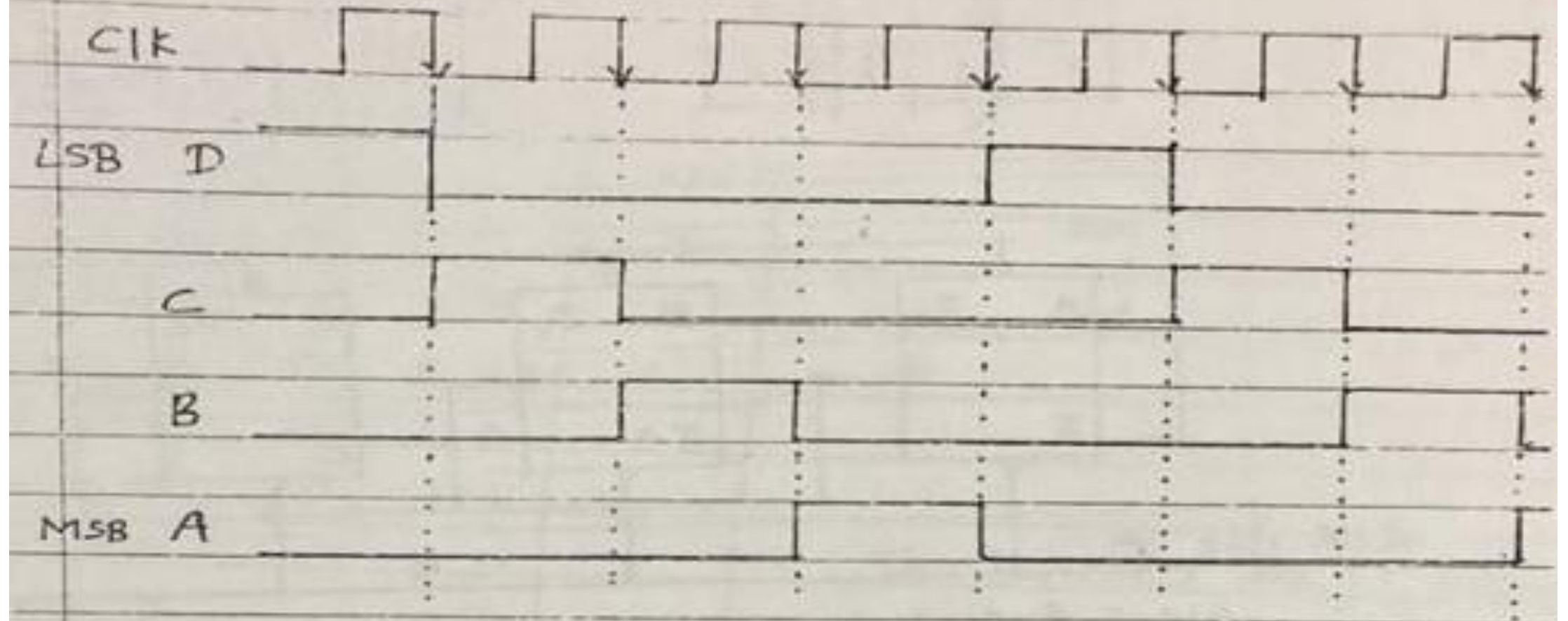




Initially when power supply is turned ON the voltage across Capacitor C will be zero. This will give o/p 0001. After $1 RC$ time constant the V_C will be $\approx 3V$. which is high level signal.

The (Clear + preset) line is at high level which will disable 3 clear pins and preset pin of LSB FF and normal operation will take place.

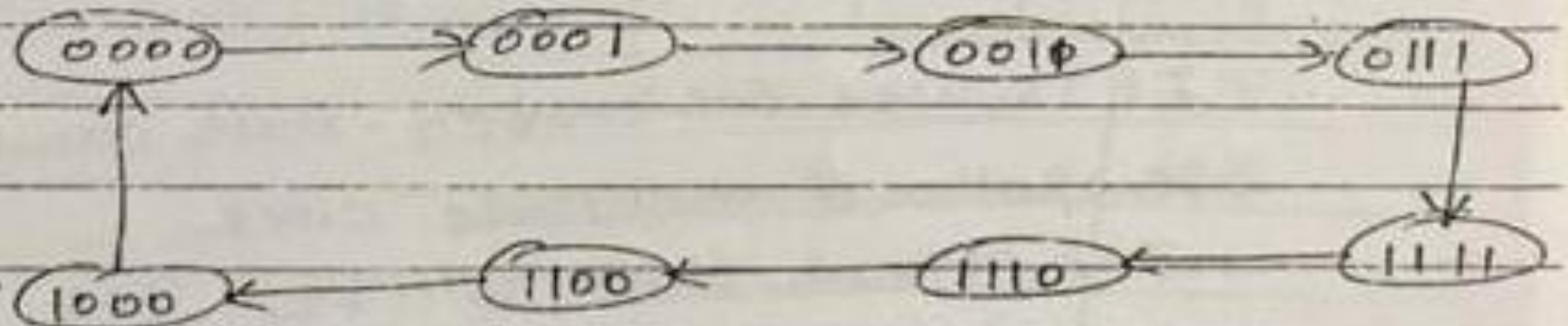
waveform of 4 bit Ring counter if -ve edge trig.
D FFs are used.



These pulses are used in stepper motors

Twisted Ring or Johnson or Moebius Counter-

In this complementary o/p of MSB FF is connecte to i/p of LSB FF. Initially all FFs are cleared by applying low level signal to clear pins of all the F. This counter is used for generation of multiple square waveform. This is divided by $2N$ counter i.e. if no of FFs are 4, then different states are 8.



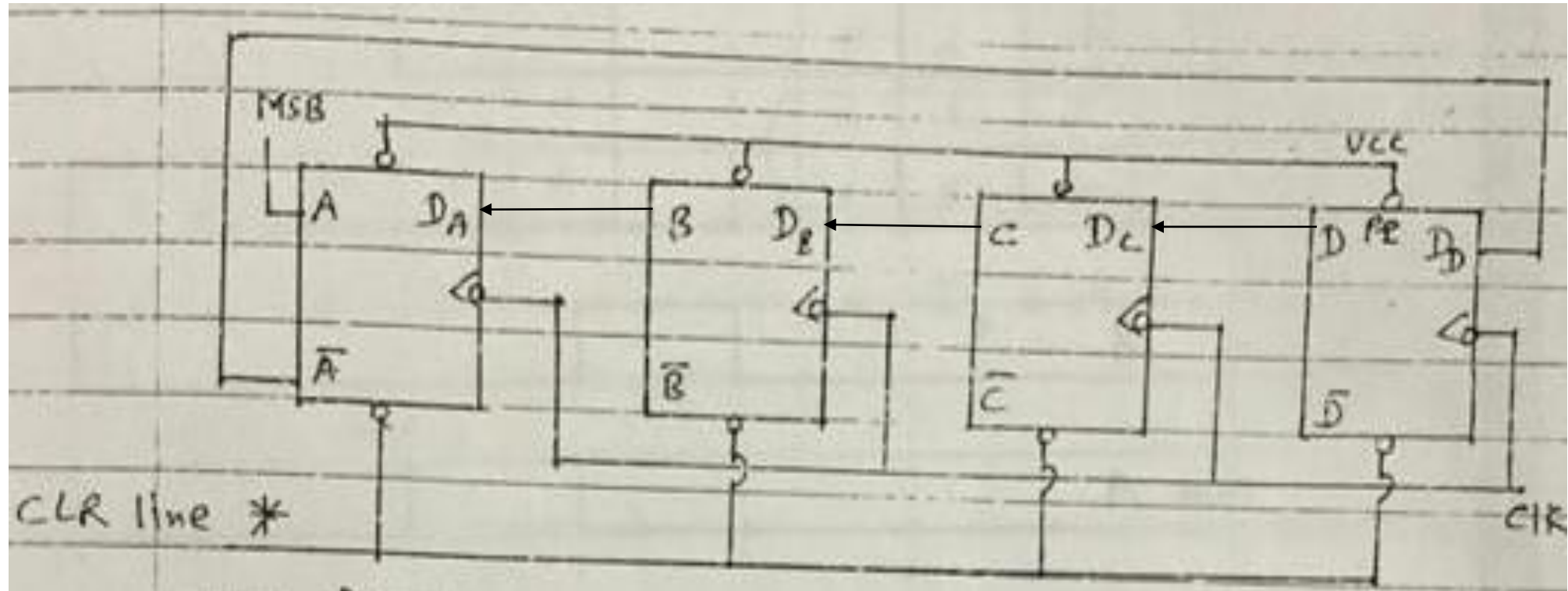
P/S				N/S				FF I/PS			
A	B	C	D	A	B	C	D	D _A	D _B	D _C	D _D
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	1	0	0	1	1
0	0	1	1	0	1	1	1	0	1	1	1
0	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	0	1	1	1	0
1	1	1	0	1	1	0	0	1	1	0	0
1	1	0	0	1	0	0	0	1	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0

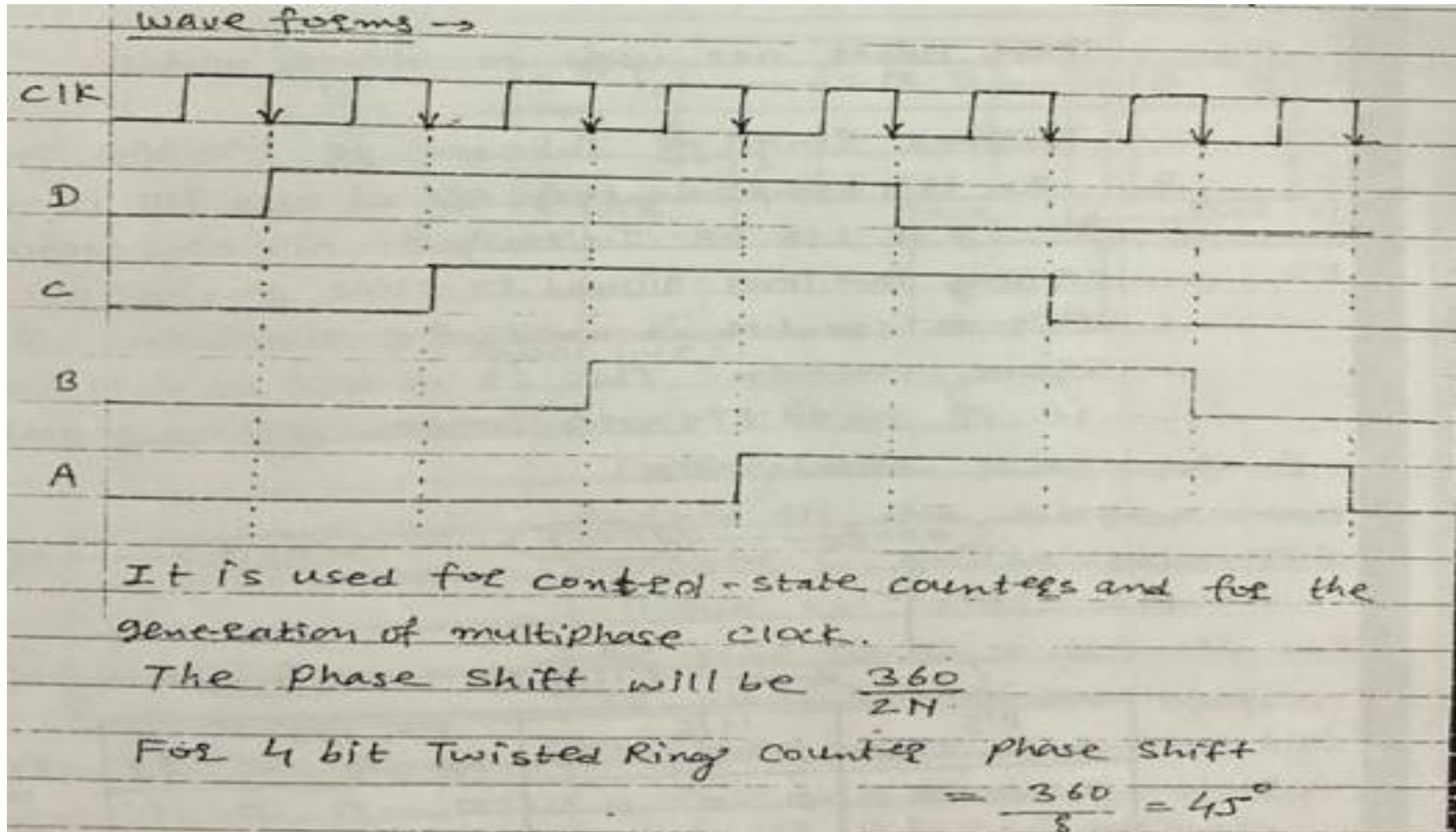
	$\bar{C}D$	$\bar{C}\bar{D}$	CD	$C\bar{D}$	
$\bar{A}\bar{B}$	0	0	0	1	
$\bar{A}B$	1	1	0	1	
AB	0	1	0	0	
$A\bar{B}$	0	1	1	1	$D_A = B$

	$\bar{C}D$	$\bar{C}\bar{D}$	CD	$C\bar{D}$	
$\bar{A}\bar{B}$	0	0	1	1	
$\bar{A}B$	1	1	1	1	
AB	0	1	1	1	
$A\bar{B}$	0	1	1	0	$D_B = C$

	$\bar{C}D$	$\bar{C}\bar{D}$	CD	$C\bar{D}$	
$\bar{A}\bar{B}$	0	1	1	1	
$\bar{A}B$	1	1	1	1	
AB	0	1	1	0	
$A\bar{B}$	0	1	1	1	$D_C = D$

	$\bar{C}D$	$\bar{C}\bar{D}$	CD	$C\bar{D}$	
$\bar{A}\bar{B}$	1	1	1	1	
$\bar{A}B$	1	1	1	1	
AB	0	1	0	0	
$A\bar{B}$	0	1	1	1	$D_D = \bar{A}$





Registers

Registers are classified depending upon the way in which data are entered or retrieved

There are 4 modes of operation:

- SISO –Serial In Serial Out
- SIPO –Serial In Parallel Out
- PISO –Parallel In Serial Out
- PIPO –Parallel In Parallel Out

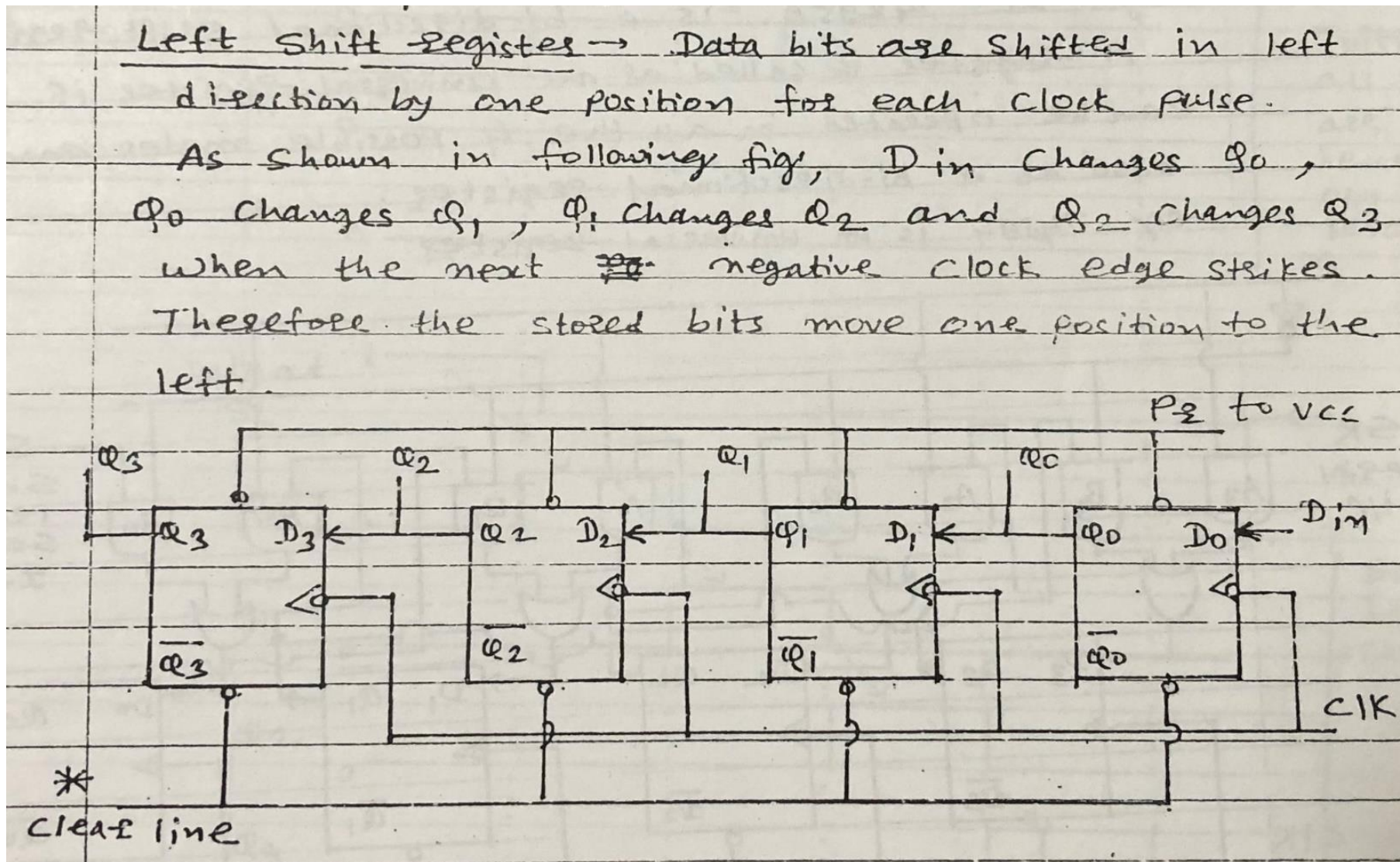
Registers are designed using D or SR or JK flip flops and are also available in MSI (Medium Scale Integration) devices.

Registers in which data is entered and / or taken out in serial way are called as shift registers. All the shift registers move stored bits in left or right direction. The bit shifting is required in certain arithmetic and logic operations used in microprocessor.

Left Shift Register

- Data bits are shifted in left direction by one position for each clock pulse.
- Din is applied as input to LSB FF and each Q output acts as input for next FF on left side and changes its content. When negative edge arrives, the stored bits move by one position in left direction.

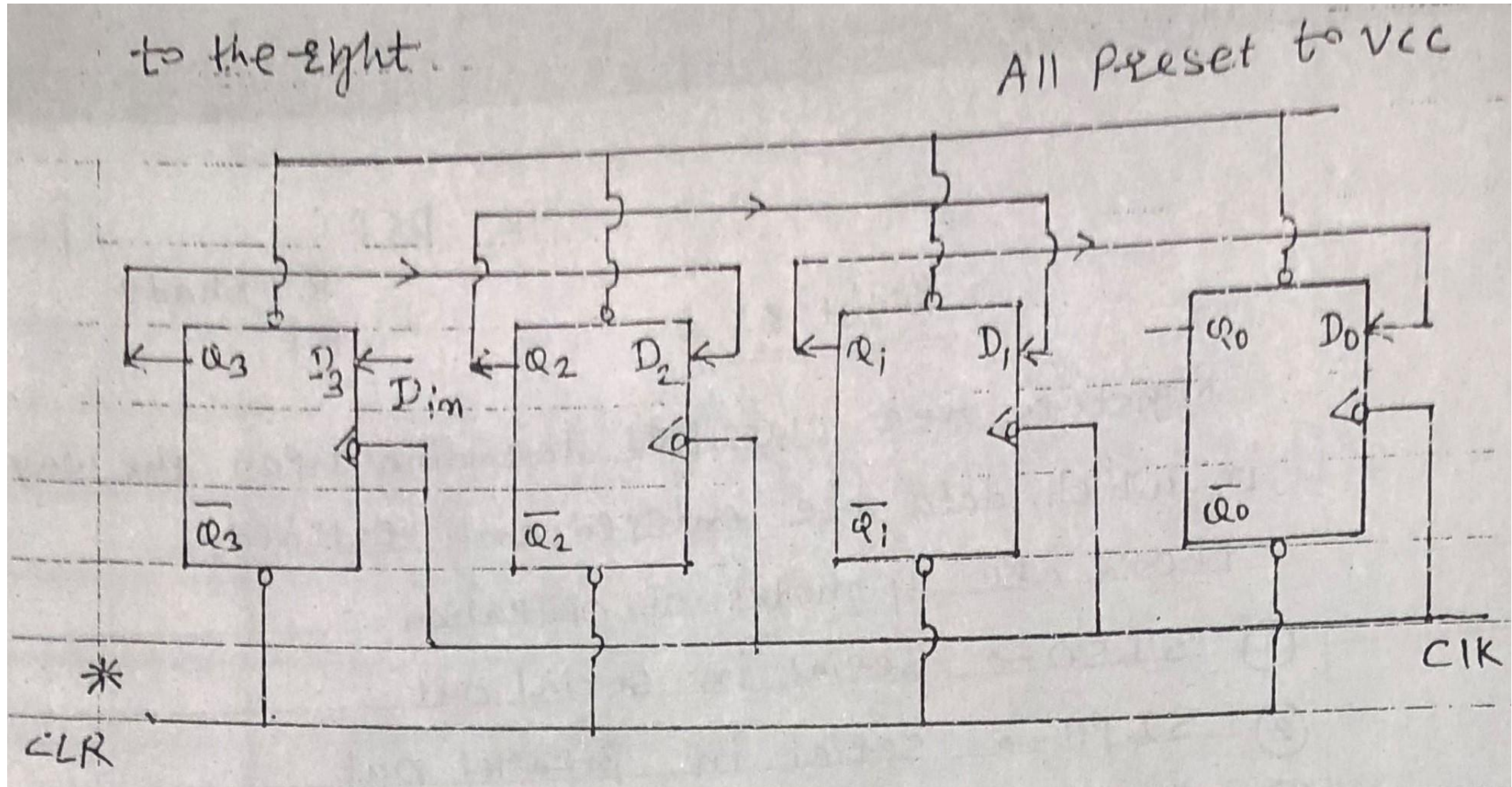
Left Shift Register



Right Shift Register

- Data bits are shifted in right direction by one position for each clock pulse as shown in the next fig.
- Din is applied as input to MSB FF and each Q output acts as input for preceding FF and changes its content. When negative edge arrives, the stored bits move by one position in right direction.

Right Shift Register



Bi-directional Shift Register

- There are applications in which it is required to shift data to the left or right. For ex: A binary number can be divided by 2 by shifting it one position to the right. Similarly, a number can be multiplied by 2 by shifting it one stage to the left.
- In Bi-directional shift register, data can be shifted from left to right as well as in the reverse direction using the mode control line.
- IC 74295A is a bi-directional shift register
- A register is called as an universal register if it is operated in all the 4 possible modes and also as a bi-directional register.
- IC 74194 is an universal register.

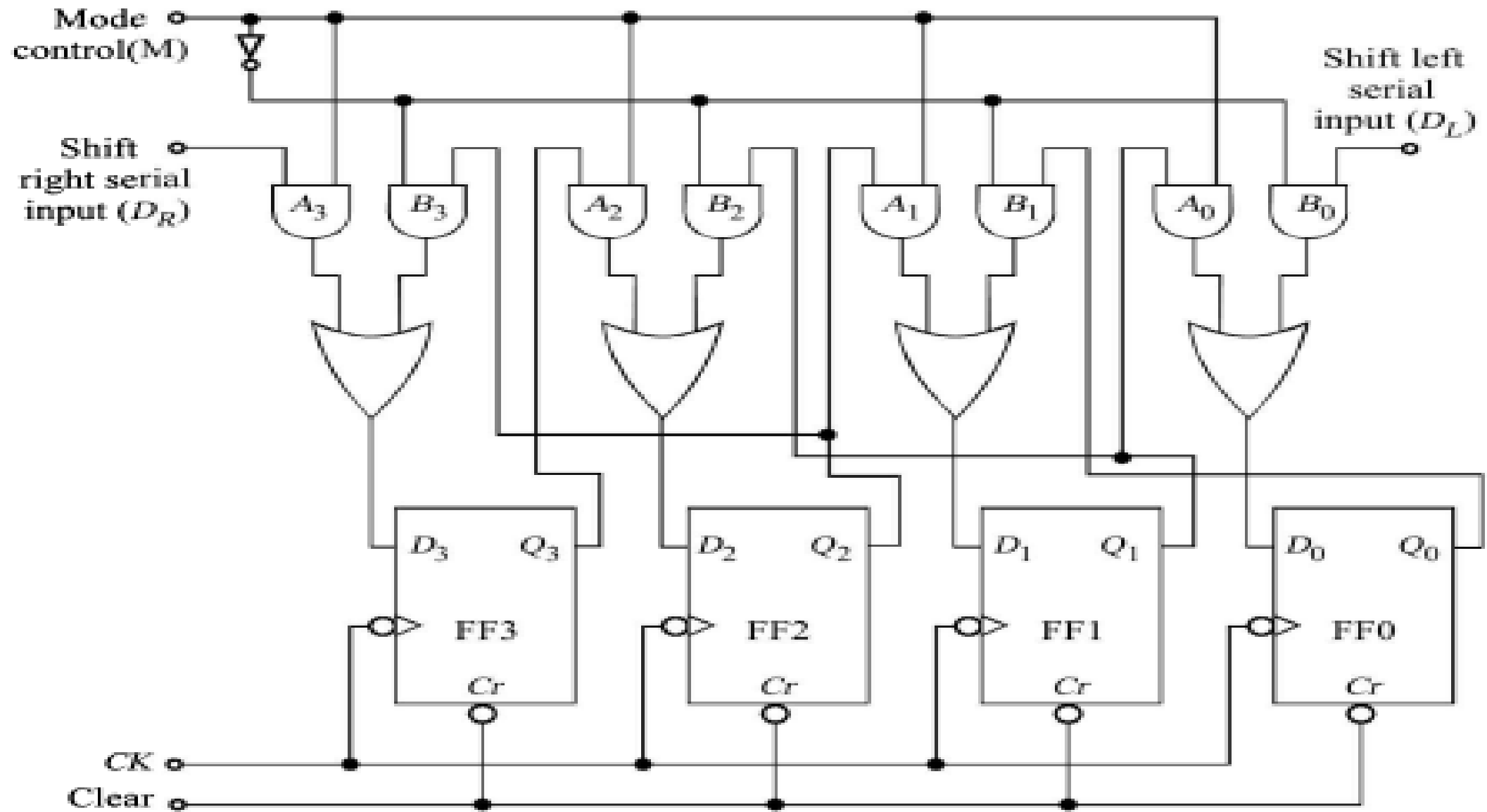


Fig. 8.4 *A 4-bit Bi-directional Shift Register*

When the mode control **M = 1**, all the **A** AND gates are enabled and the data at SR is shifted to the right when clock pulses are applied.

On the other hand, when the mode control **M = 0**, A AND gates are disabled and **B AND** gates are enabled allowing the data at SL to be shifted to the left.